MIL-T-52951(ME)
17 April 1978

MILITARY SPECIFICATION

TRUCK, CONTAINER HANDLER, WITH TOP LIFT ATTACHMENTS, DIESEL ENGINE DRIVEN, PNEUMATIC TIRED, FOUR WHEEL DRIVEN, ARTICULATED FRAME STEER, ROUGH TERRAIN; 50,000 POUND CAPACITY AT 48 INCH LOAD CENTER

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for diesel engine driven, pneumatic tired, four wheel drive, articulated frame steer, rough terrain container handler trucks, with front mounted hydraulically or electrically operated container top handler(s).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

GG-P-455	- Plates, and Foils, Photographic (Photosensitive Anodized Aluminum).
TT-C-494	- Coating Compound, Bituminous, Solvent Type, Acid Resistant.
PPP-B-636	- Boxes, Shipping, Fiberboard.

FSC 3930

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to : US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the and of this document or by letter.

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MIL-V-173	- Varnish, Moisture-and-Fungus Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-1283	- Can, Gasoline, Military, 5-Gallon.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-C-21567	- Compound, Silicone, Soft Film.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty Single Package.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-A-52363	- Air Cleaners, Intake: Dry Type (For internal Combustion Engine).
MIL-E-52649	- Engine Cold Starting Aids, Ether Fuel Primers.
MIL-F-52723	- Filter Element, Hydraulic, Disposable.
MIL-E-52798	- Enamel, Alkyd, Camouflage.
MIL-E-52835	- Enamel, Modified Alkyd, Camouflage, Lusterless.
MIL-S-62164	- Semitrailer, Low Bad, Heavy Equipment Transporter, 60 Ton M747.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-162	- Materials Handling. Equipment: Preparation for Shipment, Storage, Cyclic Maintenance, Routine Testing and Processing.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1366	- Packaging, Handling, Storage and Transportation System Dimensional and Weight Constraints, Definition of.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.

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MS51117	- Pintle Assembly, Towing, 100,000 Lbs. Capacity, Manual Release.
MS52131	- Connector, Plug, Electrical Inter-vehicle Power Cable.
MS500004	- Pad Eye Vehicular Towing Lug.
MS500048	- Towbar, Motor Vehicle: Light, Medium, and Heavy Duty.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B93.28 - Method for Calibration of Liquid Automatic Particle Counters.
MH11.3 - Load Handling Symbols for Powered Industrial Trucks.
Z26.1 - Safety Glazing Materials for Glazing Motor Vehicles Operating on Land Highways.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Welding Qualifications, Section IX.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

INTERNATIONAL STANDARDS ORGANIZATION (ISO)

ISO 668 - Freight Containers - External Dimensions and Ratings.
ISO 1161 - Specifications of Corner Fittings for Series I Freight Containers.

(Applications for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, 2501 N.W. Seventh Street, Miami FL 33125.)

INDUSTRIAL FASTENERS INSTITUTE (IFI)

Standard 114 - Standard for Break Mandrel Blind Rivets.

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(Application for copies should be addressed to the Industrial Fasteners Institute, 1517 Terminal Tower, Cleveland, OH 44113.)

NATIONAL BUREAU OF STANDARDS (NBS)

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

NATIONAL FIRE PROTECTION ASSOCIATION

National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 470 Altantic Avenue, Boston, MA 02210.)

NATIONAL FLUID POWER ASSOCIATION

T.3.9.18 - 1976 - Method of Establishing the Flow Degradation of Hydraulic Fluid Power Pumps When Exposed to Particulate Contaminant.

(Application for copies should be addressed to National Fluid Power Association, 3333 N. Mayfair Road, Milwaukees WI 53222.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

SOUTHERN CALIFORNIA AIR POLLUTION CONTROL DISTRICT

Rule 102 - Photochemically Reactive Solvents.

(Application for copies should be addressed to the Southern California Air Pollution Control District, 434 South San Pedro Street, Los Angeles, CA 90013.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

UNITED STATES DEPARTMENT OF AGRICULTURE

Forest Service Standard 5100-1.

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(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Description. The container handling truck (hereinafter called truck) shall be a four wheel drive, pneumatic tired, diesel engine driven front loading truck, furnished with container tophandler(s) as specified herein. The truck shall operate as specified herein with ISO designation 1A or IC containers or Sealand Containers loaded to a gross weight as specified. Unless otherwise specified herein, the term "truck" shall be construed to mean the container handling truck equipped with the 40 foot tophandler specified herein.

3.1.1 Rated load. Rated load for the container handling truck shall be 8 foot high ISO 668 and ISO 1161 designation 1A and 1C containers, and a Sealand Container, each loaded to a gross weight of 50,000 pounds plus 0, or minus 500 pounds. Horizontal load center (forklift longitudinal load center) shall be at the center of the container plus 0 or minus 2 inches. Vertical load center shall be at the center of the container plus or minus 2 feet. Lateral load center shall be at the center of the container plus or minus 5 percent of container length.

3.1.1.1 Carry position. Carry position for the container handling truck shall be defined as the container being carried at full back tilt and 12 inches above the ground measured to the lowest edge of the container. The oscillation and the side shift functions shall be centered.

3.1.2 Transportability. The truck shall withstand without damage highway transport, marine, and rail transport as specified herein.

3.1.2.1 Highway transport. The truck shall be transportable on semi-trailer(s) conforming to MIL-S-62164. Limited disassembly of the truck without cutting or welding is acceptable to meet height, width, length, and weight requirements specified herein. Disassembly and loading on semi-trailer(s) shall be accomplished in not more than 16 hours, and reassembly of the truck shall be accomplished in not more than 16 hours, using available manpower, two 20 ton cranes, and 10,000 pound capacity forklifts as required. Slinging and tiedown provisions conforming to MIL-STD-209 shall be furnished on each assembly, (except tires and wheels), required to be lifted by crane on to the semi-trailer. The limitations for highway transport are:

Height - when mounted on the 44 inch high semi-trailer bed the maximum height above ground shall not exceed 174 inches.

Width - The maximum width for highway transport shall not exceed 150 inches.

Length - The maximum length for highway transport shall be compatible with the semi-trailer with overhang permitted.

Weight - The maximum payload weight for highway transport shall not exceed 72,000 pounds.

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3.1.2.2 Marine transport. The truck without any tophandler shall be driveable onto and transportable on the LCU-1466, and LCU-1646 lighters. Dimensional compatibility of the truck with the lighters shall be determined from the data specified in MIL-STD-1366.

3.1.2.3 Rail transport. The truck, with mast and tophandler removed, shall be transportable by rail and shall have no permanent deformation or damage when tested in accordance with 4.6.2.13.

3.2 First articles.

3.2.1 First article (preproduction model). Unless otherwise specified (see 6.2), the contractor shall furnish two or more trucks as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods and choice of design detail will produce trucks that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4, and, unless otherwise specified herein, all examination and tests shall be conducted by the contractor, subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction model examination and tests.

3.2.2 First article (initial production). Unless otherwise specified (see 6.2), the contractor shall furnished two or more trucks for inspection as specified in 4.4.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.5).

3.3.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless suitably protected against electrolytic corrosion. Dissimilar metals and methods of protecting them are defined and described in MIL-STD-889.

3.3.2 Screw threads. Screw threads shall be in accordance with NBS Handbook H28.

3.3.3 Tire material. Tires shall not exceed 8 calendar quarters in age from date of cure of the rubber to date of acceptance of the truck by the Government.

3.4 Environmental and use requirements.

3.4.1 Operating conditions. The truck shall withstand the usage encountered in military operations without damage, permanent deformation or failure of components. The military operations include handling and stacking of the 8 foot wide family of commercial and military owned ISO designation 1A and 1C containers and Sealand containers weighing 50,000 pounds or less, and the loading and unloading of containers from flatbed trailers and rail cars. Operations will be over rough terrain, including unprepared, unstabilized

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surfaces such as beaches, snow, mud, and cross country, where high flotation is needed.

3.4.2 Operating temperatures. The engine shall start without preheating within 5 minutes and the truck shall operate as specified herein within 15 minutes after engine start in any ambient temperature in the range of minus 25° F to plus 125° F.

3.4.3 Storage temperature. The truck shall not be damaged by storage under the conditions indicated in Table I.

Table I. Storage

Air Temperature	Time Period
Plus 160° F	1 hour daily.
Plus 155° F	4 hours daily.
Minus 30° F	6 hours daily (nonwinterized)

3.4.4 Rain. The truck shall start within 5 minutes and operate as specified herein when tested as specified in 4.6.2.28,

(a) without malfunction of any electrical component, gage, or instrument;
(b) without detrimental leakage of water in the air intake filter;
(c) without leakage of water into any drive component instruments, gages, or fuel tank; and,
(d) without detrimental leakage into the cab.

3.5 Safety. All exposed parts subject to high operating temperatures or energized electrically shall be located, insulated, or guarded so as to prevent hazards to operating personnel. All rotating and reciprocating parts shall be guarded when such parts are exposed to contact by the operator. Protective devices shall not impair the operating functions. Steps and platforms for entering and servicing the truck shall have antiskid surfaces. A warning notice shall be marked near the steering hinge on both aides of the truck stating "WARNING, NO CLEARANCE FOR A PERSON STANDING IN THIS AREA", or words to that effect (see 3.31.1). Nonfunctional sharp edges, projecting points, and excessive length of fasteners shall be avoided.

3.5.1 Roll Over Protective Structure (ROPS) and Falling Objects Protective Structure (FOPS). The truck shall be equipped with a complete ROPS and FOPS structure conforming to SAE J1040 and SAE J1043. Installation and removal of the ROPS and FOPS shall be accomplished without cutting or welding. The ROPS and FOPS shall be free from sharp edges, particularly at any point where a sling may come in contact during lifting (see 3.30).

3.5.2 Seatbelt. An adjustable, weather resistant seatbelt shall be provided with a corrosion resistant, metal-to-metal, quick release buckle conforming to SAE J386, Type I and shall be anchored to the seat in accordance with SAE J787.

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3.5.3 Backup alarm. A backup alarm conforming to SAE J994 Type B shall be furnished. The backup alarm signal shall be audible above, and distinguishable from, the surrounding vehicle noise level.

3.6 Human factors. The characteristics of the truck shall provide for operation by personnel ranging from the small man clothed, through the large man, arctic clothed, in accordance with SAE J833 and SAE J925.

3.6.1 Noise limits. The noise level of the truck (excluding horns) shall not exceed:

(a) 85 dB(A) at the operator's station when tested, in accordance with 4.6.2.27.1.
(b) 85 dB(A) when tested in accordance with 4.6.2.27.2.

3.6.2 Noise hazard. The precautions of MIL-STD-1474A (see 6.6), and noise hazard warning signs shall be provided if the noise level exceeds 85 dB(A) when tested in accordance with 4.6.2.27.3.

3.7 Maintainability. Provision shall be made for adjustment, servicing, or replacement of parts and components. When access openings are used on the exterior of the truck, the edge of the opening shall be smooth and shall be provided with a removable or hinged cover whenever it is required. Dimensions of hand or arm access openings shall be in accordance with SAE J925 for arctic clothing. When tested as specified in 4.6.2.35, the truck shall have a Maintenance Ratio (MR) of not more than 0.35. The MR is defined as the ratio of total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components are excluded. Scheduled maintenance shall be required not more often than every 50 hours, except for before and after operation checks. Each operation in the following maintenance operations list shall be accomplished by one man (except where otherwise specified) in not more than the time specified, using common tools and special tools, if any, furnished with the truck:

(a) Remove, replace, and adjust all engine driven belts - 1-1/2 hours.
(b) Remove and replace alternator - 1 hour.
(c) Remove and replace all hydraulic system filters, screens, and strainers - 4 hours.
(d) Drain engine coolant, remove and replace engine coolant system hoses and thermostat, and refill system - 4 hours.
(e) Drain engine lubricating oil, remove, and replace oil filter elements and refill crankcase - 1 hour.
(f) Remove and replace fuel filter elements - 1/2 hour.
(g) Disconnect battery cables, remove and replace batteries, reconnect battery cables - 2 men - 2 hours.
(h) Drain torque converter oil and transmission oil, remove and replace all transmission filter elements and strainers, and refill converter and transmission (if applicable) - 3 hours.
(i) Remove and replace starter - 1 hour.
(j) Bleed and adjust brakes and refill master cylinder (if applicable - 2 men - 1 hour.
(k) Remove and replace floodlight and taillight bulbs (not including blackout lights) - 1/4 hour each bulb.

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(l) Disassemble as required to service air induction system including the cleaning of and replacement of the filter element - 2 hours.

3.8 Reliability. The specified Mean-Time-Between-Failure (MTBF) shall be 100 hours when tested in accordance with 4.6.2.35 at a cycle rate of not less than 3.50 cycles per hour or more than 3.75 cycles per hour.

3.9 Performance. The truck shall conform to the requirements specified herein under all conditions specified without malfunction, damage, or permanent deformation. There shall be no evidence of accelerated wear or failures when operated under service conditions specified herein.

3.9.1 Stability.

3.9.1.1 Static. With the truck stationary on a hard level surface, fueled to not more than one-quarter capacity and positioned with steering straight ahead, the rear wheels shall not clear the ground when:

(a) 1.6 times rated load is suspended from the 40 foot tophandler at a minimum 4 foot lift height and mast vertical.
(b) Rated load is tilted to maximum forward tilt from maximum rearward tilt at maximum lift height with engine at low idle speed for hydraulic mechanical drive and engine at operating speed for electric drive.

3.9.1.2 Dynamic. The truck shall negotiate a 15 percent slope with and without rated load at maximum side shift position and oscillation centered in full circle operation at maximum steer angles in both directions without any wheel leaving the ground. If the truck is furnished with a steering angle limit override which is engaged by the operator by activation of a control other than the steering-wheel, then the 15 percent slope full turn may be conducted without overriding the normal steering limit stop. The truck, while descending 25 percent grade in the forward direction with rated load in the carry position at not less than 2 mph, shall be capable of being stopped and held on the slope by the operator when applying the service brake or regenerative brake, without the rear wheels leaving the ground.

3.9.2 Vehicle flotation. The truck shall have a Flotation Index (FI) of not more than 35 with and without rated load. The FI shall be calculated using the following formula:

$$\text{FI} = \text{Antilog}\left[0.211 - 0.350C + 0.00710P - \frac{0.0752\left(P_F - P_R\right)}{P} - \frac{0.0402\left(A_F - A_R\right)}{A} + 0.0346K\right]$$

Where:

P_F = Tire pressure, front (psi)

P_R = Tire pressure, rear (psi)

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P = Tire pressure, average = 1/2 (P_F + P_R)

A_F = Axle load, front (lbs)

A_R = Axle load, rear (lbs)

A = Gross vehicle weight = A_F + A_R

K = 53.75 + 0.000139A (for nonradial tires)

K = 55.77 + 0.000032A (for radial tires)

$$C = \text{Log}\left[\frac{A}{0.607P = 1.35\left(\frac{117R}{5W + D}\right) - 4.93}\right]$$

R = Tire ply rating
W = Tire width (in. - as cited in tire size)
D = Rim diameter (in. - as cited in tire size)

The FI shall be determined with the truck fully fueled and serviced with lubricants. For the purpose of calculating FI, the tire inflation pressures may be reduced by not more than 30 percent of the recommended pressure published by the tire manufacturer for the truck's maximum tire load at a 2-1/2 mph rating.

3.9.3 Longitudinal gradeability. The truck shall ascend a 25 percent grade in forward gear range at a speed of not less than 2 mph with rated load in carry position.

3.9.4 Travel speed. The truck shall be capable of travel speeds on a level surface of not less than 9 mph forward without rated load and not less than 9 mph in reverse with rated load.

3.9.5 Fording. The truck shall operate in seawater to a depth of 60 inches including waves. Truck components shall be protected from damage, malfunction, and corrosion due to immersion followed by drying for those components below the fording level, and due to splash and spray for those components above the fording level. When tested as specified in 4.6.2.30 for a minimum of 2.5 hours the truck shall operate during and after fording operation without malfunction or damage from corrosion or salt. The volumetric water content of lubricants and hydraulic fluid shall not exceed 1.5 percent, and there shall be no measurable increase of water in fuel at the conclusion of fording tests.

3.9.6 Lifting speed. Speed of lift with rated capacity load shall be not less than 30 feet per minute over the entire distance from ground level to maximum lift height.

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3.9.7 Lowering speeds. Speed of lowering tophandler (and forks when furnished) with rated load shall be not more than 80 feet per minute over the entire distance from maximum lift height to ground level. Speed of lowering unloaded tophandler (and forks when furnished) shall be not less than 35 feet per minute over the entire distance from maximum lift height to lowest level.

3.9.8 Overall height. Overall height of the unloaded truck including mast without tophandler shall not exceed 190 inches. If the collapsed mast height exceeds 190 inches, the overall height of 190 inches maximum may be achieved by providing forward tilt in excess of that specified in 3.9.10, providing a positive stop, (such as a limit switch), is provided to limit the forward tilt for truck operation to that specified in 3.9.10 and an override control is provided to release the forward tilt stop. The override control shall not be located in the area of the container controls. The collapsed mast height shall not exceed 228 inches (19 feet) when the mast is vertical.

3.9.9 Lift height and low engagement height. The truck shall be capable of lifting the rated load in the 8 foot high containers specified, from the ground to a minimum of 113 inches, measured to the bottom of each container when the container is horizontal. The truck shall be capable of engaging the specified containers when the bottom of the container is a minimum of 12 inches below the ground level of the wheels. If lift is provided by winch and cable, positive means shall be provided to lock the winch and prevent load drift.

3.9.10 Upright tilt. When unloaded, the truck mast shall have 4.5 to 6 degrees forward tilt and 8 to 13 degrees rearward tilt. Positive means shall be provided to prevent cavitation of the hydraulic tilt cylinders (if utilized) at a minimum of 120° F oil temperature when tilting rated load to full forward tilt at engine low idle speed. If tilt is provided by electric drive, positive means shall be provided to maintain tilt position without drift. With rated load at maximum height, and engine at governed speed, the maximum time to tilt from full rear to full forward tilt shall be no more than 10 seconds, and tilt from full forward to full rearward tilt shall be no more than 10 seconds.

3.9.11 Side shift. The truck shall be capable of side shifting rated load in the specified containers a minimum of 12 inches each side of center (24 inches total). The truck, when positioned on a 15 percent side slope, shall be capable of siddshifting rated load uphill when the load is parallel to the side slope.

3.9.12 Drift. The hydraulic lift assembly shall be capable of holding the rated load at maximum height not less than 20 minutes with not more than 1 inch of vertical drift and not more than 0.5 degree of rotational drift from the vertical. Specified drift rates shall be measured when hydraulic oil temperature at the inlet to each cylinder is not less than 120° F.

3.9.13 Oscillation. The truck shall be capable of engaging and disengaging rated loads when the containers are positioned such that one end of the container is 18 inches higher than the other end.

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3.9.14 Reach. The truck shall be capable of engaging and depositing rated loads a minimum of 18 inches from the leading edge of a minimum 8 foot high platform. When carrying rated load in the horizontal position and in carry position, there shall be a minimum of 3 inches clearance between the carriage and the container wall nearest the carriage.

3.9.15 Ground clearance. The truck shall have an approach angle and a departure angle of not less than 20 degrees. The underclearance beneath the mast shall be not less than 12 inches when the tophandler is raised 12 inches and backtilted. The underclearance beneath the truck, other than the mast, shall be not less than 18 inches. Hydraulic fittings, hoses, tubing, and linkages shall not be the lowest portion of the truck and shall be protected by structural members from striking obstacles.

3.9.16 High speed. The unloaded truck shall be capable of 4 hours of high speed operation when tested as specified in 4.6.2.17 without malfunction, damage, or permanent deformation, and without exceeding specified temperature limits for the cooling system (see 3.10.2.6), engine oil (see 3.10), transmission oil (see 3.11.1), universal joints (see 3.11.1), and hydraulic system (see 3.12.1).

3.9.17 Endurance. The truck shall be capable of completing 1400 cycles in not more than 400 hours of truck operations when tested as specified in 4.6.2.34 without malfunction, damage, or permanent deformation, and without exceeding specified temperature limits for the cooling system (see 3.10.2,6), engine oil (see 3.10), transmission oil (see 3.11.1) when transmission is furnished, universal joints (see 3.11.1) when furnished, or hydraulic system (see 3.12.1).

3.9.18 Curb clearance circle. The curb clearance circle when measured in accordance with Field Test Procedure in SAE J695 shall not be greater than 70 feet when carrying rated load in a 20 foot container in the carry position for both the forward and reverse directions.

3.9.19 Slew. The truck shall be capable of engaging and lifting the horizontal rated load in the specified containers when the length of the container is at a minimum of 3.5 degrees angle to the front of the carriage. Articulation of the steering system is acceptable to meet this requirement.

3.10 Diesel engine. For the purpose of selecting the engine, the truck shall be classified as a Class II end item in accordance with MIL-STD-1410. The engine, including all systems, components, and accessories, shall conform to MIL-STD-1410, unless otherwise specified herein. All engine accessories shall be furnished by or approved by the engine manufacturer for this application. The engine shall be capable of operating on fuels and lubricating oils as specified in MIL-STD-1410. Unless otherwise specified herein, the engine shall meet all operational requirements and conditions specified herein without exceeding any of the operating limits specified in Table I of MIL-STD-1410, or the engine manufacturers published recommended operating limits. Emergency manual shutdown provisions shall be furnished when recommended by the engine manufacturer.

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3.10.1 Power and speed rating. The power and speed rating of the engine shall be such that operation of the truck under any of the operational requirements and conditions specified herein will not require horsepower in excess of the intermittent horsepower rating of the engine for the applicable governed speed as established by the engine manufacturer.

3.10.2 Engine accessories.

3.10.2.1 Starting priming system. If an auxiliary fluid priming system is required to meet the requirements of 3.4.2 it shall be operable from the operator's seat and conform to MIL-E-52649, Type III. The size of the primer shall be in accordance with the engine manufacturer's recommendations. The primer shall be electromechanically operated and shall be furnished with a temperature control as specified in MIL-E-52649.

3.10.2.2 Alternator. A heavy-duty nominal 24 volt alternator in accordance with SAE J180 with reverse polarity protection and double lug mounting, shall be provided. The alternator shall be not less than 50 amp capacity and shall provide charge at engine low idle speed. The alternator shall be driven by the gear train, cog belt, or single or matched multiple V-belts, conforming to SAE J636 and SAE J637. The regulator shall be integral with the alternator.

3.10.2.3 Starting system. The starting system shall be 24 volts and shall include, as a minimum, a starting motor solenoid, an auxiliary relay, a starting motor, necessary wiring and cable, and starting motor switch. A means shall be furnished to prevent energizing the starter motor whenever the engine is running. The truck shall be provided with either an inter-lock in the starting system or other means to prevent energizing the starter motor except when the drive train is in neutral or a mechanism to prevent the machine from moving if the engine is started with the transmission control in other than the neutral position. The starter shall be in accordance with the engine manufacturer's recommendations. The starting motor and starter solenoid shall be splashproof.

3.10.2.4 Fuel system. The capacity of the fuel tank shall provide for not less than 10 hours operation when tested as specified in 4.6.2.34 and 4.6.2.35. The fuel tank filler cap shall be captive chained. The cap shall be marked "Diesel Fuel" with cast or embossed letters unless a data plate in accordance with 3.31.1 is furnished near the fill pipe marked with "Diesel Fuel" in not less than 1 inch letter height. The tank shall accept fuel at the rate of not less than 17 gallons per minute and shall be located so that the tank can be filled from a 5 gallon can conforming to, MIL-C-1283 by a man standing on a platform. The fuel tank shall be equipped with a drainplug at the lowest point of the tank. The drainplug shall be removable with handtools without the removal of any other component. The fuel tank shall be protected by position or otherwise from external damage. Fuel lines shall provide continuous operation at full throttle and maximum loads, including slope ascension. The fuel system shall include a fuel transfer pump, fuel strainer/water separator with provisions for draining sediment, fuel filter(s) in accordance with engine manufacturer's recommendations, and a fuel injection pump, including provisions for priming. The fuel strainer or fuel filter(s) shall be between the fuel tank and transfer pump. A shutoff valve in the fuel

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line shall be provided between the tank and the fuel filter(s), if the fuel filter(s) are located below the top of the tank.

3.10.2.5 Air cleaner. The engine shall be furnished with a heavy-duty, dry type, air cleaner in accordance with MIL-A-52363, Class II or Class III. The cleaner shall be of the dual element or two stage type and shall include a dust scavenger. The capacity of the air cleaner shall be in accordance with the engine manufacturer's recommendations. An air cleaner intake restriction indicator shall be provided and shall be visible to the operator from the operator's seat.

3.10.2.6 Cooling system. The cooling system shall be of the closed pressure type, incorporating a radiator with a radiator pressure cap, hoses, thermostat, fan, and circulating pump. Under all operating conditions specified herein, the temperature differential between the cooling system temperature (radiator top tank) and ambient air temperature shall be less than the difference between the boiling point of the pressurized system and 125 ° F. Drains shall be provided to completely empty the cooling system. The cooling system fluid shall be a solution of ethylene glycol conforming to MIL-A-46153 and water.

3.10.2.6.1 Fan. The fan shall rotate on antifriction bearings. The belts shall be clear of all other components within the adjustment range. The fan belt(s) shall be in accordance with SAE J636 and J637.

3.10.2.6.2 Radiator. The radiator assembly shall consist of a core with an expansion tank and sediment tank. The radiator cap shall be accessible without removal of the engine hood. The radiator shall be mounted to prevent its being damaged due to shock and racking experienced in normal operation of the truck. The radiator shall be protected by a heavy grilled guard or by the counterweight. Hoses shall clear the truck structure and other components under all operating conditions specified herein. The cooling system shall include baffles and shroud to prevent recirculation of cooling air.

3.10.2.6.3 Engine coolant hose. All engine coolant hoses shall be SAE J20, heavy-duty type. All hose clamps shall be of the double wrap stainless steel type in accordance with SAE J536, Type F, Style 1, 2, or 4.

3.10.2.7 Governor. A governor shall be provided to limit the engine speed to not more than the published intermittent duty speed recommended by the engine manufacturer when the truck is operated under any condition specified herein.

3.10.2.8 Exhaust system. The exhaust system shall be protected against entry of rain. The back pressure of the exhaust system, including muffler, shall in no case exceed the maximum recommended by the engine manufacturer when measured approximately 2 inches beyond the header outlet at any load up to rated net continuous load. Exhaust gases shall be emitted at the rear, directed back of the truck within the plan outline of the truck. The exhaust system shall be arranged to prevent burns to the operator and maintenance personnel while working on step surfaces of the truck. The exhaust system shall be provided with a spark arrestor conforming to Forest Service Standard 5100-1, and SAE J350 for not less than 80 percent efficiency if the engine is

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not equipped with a turbocharger. The spark arrestor shall be provided with a means for cleaning out trapped particles.

3.10.2.9 Crankcase ventilation system. The crankcase ventilation system shall preclude water entering the engine during fording operations. When the crankcase vapors are directed into the induction system, means shall be provided on turbocharged engines to prevent coking of turbocharger blades. No vapors or condensation shall be deposited on or come in contact with the air cleaner element.

3.10.2.10 Oil filters. Oil filter(s) of the heavy-duty full flow or heavy-duty full flow and partial flow type shall be furnished in accordance with the engine manufacturer's recommendations. Oil lines, when applicable, to and from the filters shall be installed to minimize vibration and shall be of flexible hose.

3.11 Drive train. The drive train shall consist of all components necessary to transmit power from the engine to all four wheels. The drive train components shall conform to and shall be installed in accordance with each component manufacturer's recommendations. The drive train shall be either a hydraulic mechanical type or electric type.

3.11.1 Hydraulic mechanical drive train. The hydraulic mechanical drive train shall be furnished with a torque converter, powershift transmission, drive shafts, universal joints, driving axles and any other components necessary to drive the truck. The drive train fluid and lubricant temperatures shall not exceed 125° F above ambient and drive train universal joints shall not exceed 60° F above ambient, when the truck is tested as specified herein. Drive shafts shall be provided with a minimum of two heavy-duty universal joints and a nonbinding slip joint.

3.11.1.1 Powershift transmission. The Powershift transmission gear ratios shall be manually selected. All transmission gears shall be constant mesh. The transmission shall provide for shifting under full engine power through all successive forward and reverse transmission gear ratios. The transmission shall provide controlled application of the clutches to provide smooth shifting, limiting slippage which could damage the clutches. Oil for the transmission shall conform to MIL-L-2104, Grade 10 or MIL-L-46167. All transmission component material shall be compatible with these oils. Oil pressure for actuation of the transmission clutches, for torque converter supply, and for transmission lubrication shall be supplied by a mechanically driven pump. The transmission and torque converter fluid shall be filtered by filter(s) having replaceable elements in accordance with the transmission manufacturer's recommendations. The torque converter and transmission shall absorb maximum stall operation for a minimum of 30 seconds without damage or permanent deformation and without exceeding the fluid temperature limit of 250° F with an initial fluid temperature of not less than 157° F in an ambient temperature of not lower than 25° F. The transmission shall be provided with a disconnect device. The disconnect device shall disengage the transmission while allowing the engine to be operated throughout the entire speed range. The truck shall be capable of achieving specified lifting, lowering, and tilting rates of the rated load while the disconnect device is being used.

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3.11.2 Electric drive train. The electric drive train shall consist of a DC generator, necessary switching equipment, and DC motors and reduction gears located in each wheel. The drive train shall be capable of performing all operations specified herein without causing heating of any electrical equipment or component in excess of the manufacturer's recommended temperature rise. The drive train shall absorb maximum drawbar pull at truck stall for a minimum of 30 seconds without damage or permanent deformation and without exceeding the recommended temperature rise for the generator, switch gear, and drive motors. Thermal protectors shall not activate during the 30 second stall.

3.11.2.1 DC generator. The DC generator shall be 700 volts DC, four or six pole, two bearing, shunt wound or differentially compound wound, forced air cooled, special purpose traction type. The generator shall be drip-proofed if the generator is located above the fording depth or waterproof if located below the fording depth. Air ducts shall be rainproof. The generator shall have continuous duty rating for this application, and have insulation of Class F. The generator shall be coupled in line to the diesel engine with flexible coupling. Its shunt field excitation shall be supplied by a belt driven exciter generator. The field of the exciter generator shall be supplied by the 24 volt DC battery through a potentiometer. The DC generator shall be 2200 rpm continuous rated, variable voltage.

3.11.2.2 Switchboards. Switchboards (including motor and generator control and protective panels) shall be front connected, front wired, complete with all necessary controls, rheostats, contactors, circuit breakers, switches and other appurtenances. The AC power magnetic contactors shall be special purpose type of sufficient size to withstand full voltage severe inching. The control panels shall be arranged in a neat and orderly manner on the walls and shelves of switchgear compartments. The compartment's shall be sealed and shall be rain proofed and shall be waterproofed if any part of the compartment is below the fording depth. The compartment shall be laid out utilizing good commercial practice. All contactors shall be located above the fording depth. All wire terminations on switchgear and control devices shall be accessible. All equipment and resistors shall be adequately ventilated and readily accessible. Instruments, switches, field rheostats and similar equipment shall be identified as to the function. Switchboard wiring shall be 600 volt type AWM or DEL cable installed in such a manner that each wire is accessible and firmly supported. The wiring shall be done in a workmanlike manner and properly sized, using as reference the National Electrical Code. The wiring shall not interfere with the operation, maintenance and inspection of the switchboard components. Terminal blocks shall be provided and marked for all incoming circuits.

3.11.2.3 DC motors. The drive motors shall be Class F insulated and shall be special purpose for off highway material handling equipment. The motors shall be open type covered to make them waterproof. DC motors for travel motion shall be for continuous operation with forced air cooling, variable speed, accumulative compound wound, special purpose traction type.

3.11.3 Rear axle oscillation. The rear axle shall have a minimum of 18 degrees total free oscillation.

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3.12 Container handling and steering system power. The power for the container handling functions and steering function shall be either hydraulic or electric.

3.12.1 Hydraulic system. The hydraulic system shall consist of all hydraulic components necessary for operation of the truck including steering system and does not include brakes and power train, which are not part of the main hydraulic system. The system shall have a maximum working pressure (relief valve setting) not in excess of 3000 psi. Oil for the system shall conform to MIL-L-2104, Grade 10, or MIL-L-46167. All hydraulic component material shall be compatible with these oils. Unless otherwise specified herein, the temperature rise of the hydraulic fluid shall not exceed 100 ° F above ambient. Unless otherwise specified herein, all system components which are subject to full system pressure shall be rated to withstand a proof pressure that is equal to or greater than 1-1/2 times the maximum working pressure (1-1/2 times relief valve setting) without external leakage damage, or permanent deformation. Hydraulic fittings on pumps, valves, and cylinders shall be the O-ring straight thread type as specified in SAE J514 or 4-bolt flanged type as specified in SAE J518. The hydraulic system shall be cleaned free of foreign matter including weld slag and spatter. Hydraulic connection ports of the hydraulic system components shall be sealed by the component manufacturer. The system shall provide for lowering of rated load at the rate specified in 3.9.7 in the event of failure of or damage to hydraulic hose supplying the lift cylinder. A straight thread O-ring boss test point shall be located at the outlet of each pump or pressure inlet of each control valve for measuring pressure.

3.12.1.1 Hydraulic lines. Hydraulic lines shall consist of tubing, hose, and fittings. Hose or a combination of hose and tubing shall be used between components that are not mounted to a rigidly connected nonflexing structural member or rigidly connected subassembly. All lines shall be routed to provide the minimum number of bends, ease of maintenance, and maximum protection. Provision shall be made to prevent damage of lines due to chafing. The hose installation, such as angle of fittings and location, shall cause no stress concentration on the hose at the fitting, and the hose shall be supported, when necessary, to eliminate sagging at the fitting. All lines, other than suction lines, return lines and cylinder bleeder lines, shall be considered pressure lines. All lines shall be cleaned free of foreign matter including weld slag and spatter.

3.12.1.1.1 Pressure hose assemblies. Permanently attached type pressure hose couplings are acceptable as original equipment hose assemblies. The truck shall be compatible with hose assemblies with field attachable reuseable fittings in place of each hose assembly with permanently attached couplings when tested as specified herein. All fittings shall be 37 degree flare, female swivel or 4-bolt split flange as specified in SAE J516. Hose ID shall be limited to 1/4, 3/8, 1/2, 3/4, 1, 1-1/4, 1-1/2, and 2 inches. The hose selected shall have a working pressure that is equal to or greater than the hydraulic system maximum relief valve setting. The hose installation shall not cause the hose to be bent less than the minimum bend radius specified in SAE J517.

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3.12.1.1.2 Pressure tubing. Pressure tubing shall be in accordance with SAE J524 or SAE J525 and shall meet the requirements specified herein. Pressure tube sizes shall be limited to 1/4, 5/16, 3/8, 1/2, 5/8, 3/4, 1, 1-1/4, 1-1/2, and 2 inches. Tubing shall have a minimum burst pressure that is equal to or greater than 3.5 times the system maximum working pressure. Tubing shall be clamped by threaded fasteners. All bends shall be smooth without flattening, kinking, or wrinkling of the tube. The term "without flattening" is defined as follows:

> Throughout the bend, the tube maximum diameter shall be equal to or greater than the tube nominal diameter, and the tube maximum diameter minus the tube minimum diameter shall be not greater than 15 percent of the tube nominal diameter.

3.12.1.1.3 Pressure tube fittings. Unless otherwise specified herein pressure tube fittings shall be 37 degree flare or O-ring boss conforming to SAE J514d, or 4-bolt split flange conforming to SAE J518. Jump size tees may be used. 4-bolt split flange connections may terminate in either a flanged head that incorporates a SAE J518 mating face or a flange head that integrally incorporates four boltholes and an O-ring groove for mating to a SAE J518 port face. Pressure tube fittings that incorporate the SAE J518 connection shall contain a boss into which the tube shall be brazed or welded. Pressure tube fitting sizes shall be limited to 1/4, 5/16, 3/8, 1/2, 5/8, 3/4, 1, 1-1/2, 1-3/4, and 2 inch. Pressure tube fittings shall have a minimum burst pressure that is equal to or greater than 3.5 times the maximum system working pressure.

3.12.1.1.4 Suction lines. The pump suction line (pump inlet line) shall include provisions for flexibility when there is vibration or movement between the reservoir and the pump. The suction line shall be sized so that the minimum pump inlet pressure does not fall below atmosphere by more than 7 inches of mercury vacuum at 150° F plus or minus 10° F oil temperature and at governed engine speed. The suction line installation shall include no stress concentration at the end fittings and no kinks. Suction hose shall conform to SAE J517, 100R4 or shall be oil resistant hose. Suction hose clamps shall be of the double wrap stainless steel type in accordance with SAE J536, Type F, Style 1, 2, or 4 or shall be in accordance with 3.12.1.1

3.12.1.1.5 Return lines. Return lines shall be in accordance with SAE J517 except 100R7 type hose is not acceptable.

3.12.1.1.6 Quick disconnects. Quick-disconnect couplings shall connect to lines 1 inch and smaller with an O-ring boss conforming to SAE J514. The couplings shall connect to lines larger than 1 inch with the SAE J518, 4-bolt, split-flange face, 3,000 psi. The coupling halves shall be restrained to prevent contact with the ground or any surface of the hoist or the spreader when connected and when disconnected. The coupling halves shall be identified to assure assembly of correct mating halves. Each coupling half shall be provided with captive chained cap to protect the opening when the couplings are disconnected or permanently affixed receptacles with caps to protect the opening when the couplings are disconnected. The caps or receptacles for mating coupling halves shall mate together when not in use to prevent corrosion, damage, and contamination of mating surfaces. Couplings shall not

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have any partially coupled unlocked positions in which the couplings can remain stable and permit flow. After 100 connections and disconnection, the couplings shall meet all requirements specified herein. When coupled and uncoupled with 60 psi applied, the couplings shall seal the ends of the disconnected lines at the point of disconnection and shall not permit external fluid loss exceeding 1/2 fluid ounce. At full-line flow, maximum pressure drop across quick-disconnect couplings shall be 15 psi. The connected couplings disconnected halves shall with-stand 5,000 psi without evidence of leakage or rupture. The couplings shall show no evidence of leakage and shall meet all requirements specified herein.

3.12.1.2 Pump contaminant tolerance level. When the pump is tested as specified in 4.6.2.21.5, the 1000 hour contaminant tolerance profile shall not drop below or cross the profile on Figure 1.

3.12.1.3 System filtration. System filter(s) shall be furnished either in the suction line, pressure line, return line, or combination as specified in 3.12.1.3.1, 3.12.1.3.2, or 3.12.1.3.3.

3.12.1.3.1 Return line filters. Although the system filter element need not be qualified to MIL-F-52723, it shall meet the performance requirements stated therein. The filter shall maintain a full flow condition for the duration of testing as specified in 4.6.2.34 and 4.6.2.35. The filter bypass may open for system oil at a temperature less than 70° F. The filter element shall have been tested to the requirements of MIL-F-52723 at a flow rate equal to or greater than the truck(s) maximum system oil flow at; (1) maximum governed engine speed or, (2) lowering rated load at engine low idle speed, whichever is greater. All return oil shall pass through the filter except relief oil which may return directly to the reservoir. The filter shall be located so that the element may be removed for examination or replacement without spilling or draining oil from the reservoir. At the maximum system oil flow through the filter, including lowering rated load at maximum governed engine speed and an oil temperature of 150° F plus or minus 5° F, the pressure drop across a new filter assembly shall not exceed 10 psi. An automatic bypass shall be provided in the filter and shall open at not less than 10 psi greater than pressure drop measured across new filter assembly and shall permit full flow at a pressure not greater than 35 psi. The filter housing shall have a minimum proof pressure of 100 psi without permanent deformation, damage, or leakage. The filter shall be equipped with a contamination indicator which may be remotely mounted. The indicator shall be clearly readable without the removal of any components. As a minimum, the indicator shall show when the element is clean and when the filter is beginning to bypass, requiring a change of the element.

3.12.1.3.2 Suction filter. The suction filter shall conform to 3.12.1.3.1 with the following exceptions:

(a) Pressure drop across a new filter assembly shall not exceed 0.75 psi.
(b) The automatic bypass shall open at not less than 2.0 psig and shall permit full flow at not more than 3.5 psi.
(c) The filter housing shall have a proof pressure of not less than 30 psi.

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(d) Although the system filter element need not be qualified to MIL-F-52723 it shall meet the performance requirements stated therein, except that test terminal pressure drop shall be 2.5 psi and average filtration ratio (810) shall be equal to or greater than 2.2.

3.12.1.3.3 Pressure filter. The pressure filter shall conform to 3.12.1.3.1 except that proof pressure test shall be 150 percent maximum working pressure.

3.12.1.4 Reservoir. A reservoir shall be furnished to contain the hydraulic system oil. The reservoir shall include a filler opening, screen, cap, and other items required to function as specified herein. The reservoir shall be rigidly mounted to prevent vibrations that could result in leaks, loosening of fasteners, brinelling of material, or cracks. The reservoir shall have sufficient capacity to prevent air entering the system with all hydraulic pistons fully extended, and sufficient free air capacity to prevent oil being discharged through the reservoir air vent, when maximum return flow of oil (lowering rated load at maximum governed engine speed) is surged into the reservoir from the system and all cylinders are retracted. The oil level indicator (see 3.28.2) shall indicate the correct oil level when the truck is on level ground, the mast is vertical, with tophandler in down position, and the oil is at normal operating temperature. The fillercap and tube shall be located to provide for filling from a standard 5 gallon container conforming to MIL-C-1283, and to prevent entrance of contaminants throughout filling and checking operations. The reservoir filler neck shall contain a removable 30 mesh, or finer, strainer retained in place unless all oil added to the reservoir is filtered by an equal or finer filter prior to passing through the pump. The filler tube shall be not less than 1 inch inside diameter. The reservoir pump suction openings shall be located a minimum of 3/8 inch above the bottom of the reservoir. Except for reservoirs furnished with a pressure/vacuum relief breather, a breather with a pleated paper element with a minimum filtering surface area of 200 square inches shall be provided and shall filter incoming air. The breather shall preclude the entrance of water induced by rain. The reservoir shall provide a tortuous flow path from the reservoir inlet to the reservoir outlet in order to prevent a direct current between the two ports. All return line flow except the vent line shall discharge into the reservoir at a level which is below the oil level in the reservoir, when the lift cylinder is at maximum extension. A removable 50 mesh, or finer, strainer shall be installed in the suction line between the reservoir and the pump unless a suction line filter in accordance with 3.12.1.3.2 is furnished. The reservoir shall have one or more accesses for manual cleaning of the entire inside or be removable for cleaning.

3.12.1.5 Relief valve. A relief valve shall be provided to limit the pump pressure to the maximum required to perform the necessary system functions. Pressure equal to the relief valve setting shall be obtained only when a cylinder is bottomed or when the system is subjected to shock loading.

3.12.1.6 Hydraulic control valves. Mounting of the valves shall be rigid and shall permit no vibration which may cause instability of the controls. All control valve spools shall be spring centered with sufficient force to return the levers to the center position and retain them there throughout all

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operations. Spools shall not bind under all operating conditions specified herein. All exposed control valve surfaces shall show no evidence of corrosion when tested as specified herein.

3.12.1.7 Cylinders. All hydraulic system and steering cylinders shall be fabricated from either seamless steel tubing or welded and redrawn tubing. All welding and brazing to the cylinder within the zone of piston operation shall be done prior to final machine honing or rolling of the cylinder. Cylinders shall be so located as to provide for ease of maintenance and replacement. All internal cylinder ports shall be located beyond the area of piston travel. Each cylinder shall have an exclusion device to prevent foreign material and fluids from entering around the piston rod and damaging the seal and other cylinder components. All hydraulic system and steering cylinders shall perform as specified herein without buckling or bending. Each piston shall have a bearing surface concentric with the piston to carry the piston side loads against the cylinder walls. The bearing surface shall be of a material that will not scar, score, or gall the inside surface of the cylinder. All cylinder rods shall be hard chrome plated. All cylinders shall withstand a pressure that is equal to 1.5 times the system relief valve setting without evidence of rupture, permanent deformation, damage, or external leakage.

3.12.1.8 Hose reels. When hydraulic hose reels are used on the truck to maintain hose tension during mast extension, sheave wheels, guide rollers, guide blocks, or other means shall be provided to prevent the hose from becoming entangled around and riding over the reels or reel flanges.

3.12.1.9 Contamination levels for hydraulic system. When tested as specified in 4.6.2.21.3 particle counts shall not exceed 1000 particles per milliliter greater than 10 micrometers and particle counts shall not exceed 10 particles per milliliter greater than 20 micrometers.

3.12.2 Electric power system. The electric container handling system and steering system shall consist of an AC generator, necessary switching equipment, and AC motors to perform as specified herein.

3.12.2.1 AC generator. The AC generator shall be 440 volts AC, 60 Hertz, 3 phase special purpose type and rated for continuous duty at not less than 100 KW, 0.8 power factor. It shall be suitable for continuous operation, at 2200 rpm, 545 volts AC, 73 Hertz. The generator shall have an insulation Class F. Initial excitation is to be supplied by 24 volt DC batteries. A transformer rectifier system may be used to maintain operational excitation after normal AC generator output is reached thus making the AC generator a self excited generator. The excitation transformer must compensate for varying load conditions. The generator shall be dripproofed if the generator is located above the fording depth or waterproof if located below the fording depth.

3.12.2.2 AC motors. Motors shall be Class F insulated and shall be of special purpose design for off-highway material handling trucks. AC motors for steering, hoist, and tilt shall be 440 volt, 3 phase, 60 Hertz, 1,750 rpm (or less) squirrel cage, induction type. Motors shall be dripproofed if the

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motor is located above the fording depth or waterproof if located below the fording depth.

3.13 Mast, carriage, and load handlers. Mast upright(s) shall be furnished. If two stage mast is furnished it shall be telescopic roller type. Rollers shall be of the permanently lubricated for life type. Mast upright(s) and carriage shall be positioned on the truck such that a container in all positions specified herein shall not come in contact with the front tires or any other portion of the truck other than the tophandler. Mast and carriage assembly shall be removable from the truck without requiring permanent modification (such as cutting or welding) to the truck. Chains or cables shall be installed in such a manner they they cannot be displaced from the roller until they have been raised a minimum of 1/2 the chain side plate height. Adequate means shall be provided in the mast uprights to compensate by replacement or adjustment for wear which may develop between lateral rolling contacts. Carriages shall be equipped with adjustable thrust bearings to resist thrust introduced by off-center loading. Chains and chain anchors used in the lifting mechanism shall have a factor of safety of not less than 5.0 to 1 based on minimum ultimate tensile strength when truck is carrying rated load centered on the truck. Cable used in the lifting mechanism shall have a factor of safety of not less than 3.5 to 1 based on minimum ultimate tensile strength when truck is carrying rated load centered on the truck. Hydraulic lines and electrical lines between the truck and the mast shall be furnished with quick disconnects.

3.13.1 Container tophandlers. The first article (preproduction model) truck shall be furnished with three tophandlers. One tophandler shall be compatible with designation 1A freight containers (40 foot nominal length) specified in ISO 1161 specifications. One tophandler shall be compatible with designation 1C freight containers (20 foot nominal length) specified in ISO 1161 specifications. One tophandler shall be compatible with "Sea Land" containers (35 foot nominal length). All trucks shall be furnished with a tophandler compatible with designation 1C freight containers. When specified (see 6.2), trucks shall be furnished with tophandlers compatible with 1A containers. When specified (see 6.2), trucks shall be furnished with tophandlers compatible with "Sea Land" containers. The tophandlers shall be capable of being interchanged by the operator with hand tools in no more than 30 minutes. The tophandler shall safely mate, aline, couple, and lock with the containers specified. The tophandler shall safely lift, transport, and lower the coupled containers at specified rates of hoist and lower velocities. The complete tophandler shall be capable of being disconnected from the truck in a maximum of 30 minutes. The tophandler shall mount to the truck by positive connection to the carriage. Hydraulic lines and electrical lines between mast or carriage and tophandler shall be furnished with quick-disconnects.

3.13.1.1 Coupling devices. Coupling devices shall be furnished to mate the handler to the specified freight containers. Coupling devices may be air operated. One coupling device shall be provided for each corner of the container. The top lifting coupling device shall be of the twist lock type. The locations of the coupling devices shall coincide with the longitudinal, lateral, and diagonal dimensions and tolerances for the container corner fittings in accordance with ISO 1161 and Sea Land as applicable.

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3.13.1.2 Alining arms. The spreader shall be equipped with a minimum of 2 alining arms. The alining arms shall be located on the longitudinal frame rail nearest the mast assembly. An additional alining arm located on the left side 8 foot rail may be furnished. The alinement arms shall be located at the two near corners of the tophandler to coincide with the corner fittings of the containers. The minimum length below the rail shall be 18 inches. The end of the arms shall be flared for a minimum of 3 inches.

3.13.1.3 Coupler indicator lights. A minimum of three different color coupler position indicator lights located in the cab adjacent to tophandler controls or on the mast directly visible by the operator shall be furnished. Lighting of each of the indicator lights shall be easily distinguishable for for all directions of sunlight. One indicator light shall indicate all twist locks are in the unlocked position. One indicator light shall indicator all twist locks are down in the container corner fittings for both the locked and unlocked position. One indicator light shall indicate all twist locks are in the locked position.

3.13.2 Fork kit. All trucks shall be capable of accepting a fork kit including all the necessary components required for fork operations specified in 3.13.2.1. The kit shall be installed without permanent modification (such as cutting or welding) and without removal of the mast. After removal of the tophandler and installation of the fork kit, the forks shall be capable of being installed in a maximum of 1 hour and removed in a maximum of 1 hour.

3.13.2.1 Forks and fork kit. When specified (see 6.2), a fork kit including two forks shall be furnished for conventional forklift operation. Each fork shall be capable of withstanding, without permanent deformation, 150 percent of rated load at 48-inch load center measured from face of the forks. Forks shall be attached to the fork carriage in a manner which prevents lateral slippage or permanent deformation of fork carriage components. Forks shall be a maximum of 5-3/4 inches thick at the heel, a maximum of 14 inches wide, and 96 plus 0 minus 1/2 inch long. Fork thickness shall not exceed 1-1/4 inch at the tips and the end taper shall be not less than 36 inches long. Taper shall be on the underside of the forks. The top of the fork tips shall be chamfered or rounded to break corners and provide smooth edges. Each fork arm shall be stamped with its individual rated capacity in a visible area not subject to wear. The truck, when equipped with the fork kit, shall be capable of:

(a) Engaging a 50,000 pound load at 48-inch load center when the load is a minimum of 6 inches below ground level of the wheels and lifting the load to a minimum of 100 inches above the ground.
(b) Tilting a 50,000 pound load 4.5 to 6 degrees forward.
(c) Tilting a 50,000 pound load 8 to 13 degrees rearward.
(d) Power fork positioning capable of adjusting fork spacing from not more than 67 inches when together to not less than 90 inches when spread, measured at the outside edges of the forks.

3.14 Towing. The truck, with a dead engine, shall be towable at speeds up to 4 mph for up to 10 miles. Preparation of the truck for towing (such as disengaging drive train or engaging auxiliary steering) shall be accomplished

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by not more than 2 men in not more than 30 minutes without cutting or welding. The truck shall be capable of being towed around 180 degree turns in no wider than 2 times the curb clearance diameter of the truck.

3.15 Wheels and tires.

3.15.1 Wheels. The truck shall be equipped with four demountable wheels. The wheels shall be fastened to the hubs in accordance with the axle manufacturer's recommendations for off-road applications. Wheel rims shall be in accordance with the tire manufacturer's recommendations.

3.15.2 Tires. All tires shall be new and unused and of identical size and construction. The tires shall be radial, pneumatic, tubeless, furnished with nondirectional, traction or rock lug tread. Tire loadings when the truck is carrying rated load in carry position shall not exceed the load limits specified by the tire manufacturer at 2-1/2 mph. Valves and caps shall be in accordance with the TRA Yearbook. Valve caps shall be metal with elastomer innerseal. Tire ballast is not acceptable and shall not be furnished.

3.16 Steering. The steering method shall be a hinged frame (articulated) type. The steering system shall permit control of the truck with rated load while traveling at any speed up to the maximum specified travel speed. The steering system shall permit control of the unloaded truck while traveling at any speed up to the maximum capable travel speed. The truck shall operate without loss of operator control when tested as specified in 4.6.2.12. Means shall be provided to prevent wheels or tires or steering linkage from contacting any part or truck structure, other than stops, irrespective of obstacle magnitude or cramped steer angle.

3.16.1 Steering control. The steering system shall be controlled by an automotive type steering wheel not to exceed 18 inches in diameter or a control which will actuate an electric motor controlling the steering. When a steering wheel is provided, effort required to turn the steering wheel shall not exceed 15 pounds pull at the rim in either direction when the unloaded truck is stationary on a smooth, dry concrete surface with the engine at low idle speed. The steer wheels shall be capable of being turned from extreme right to extreme left or vice versa in not more than six complete turns of the hand steering wheel. There shall be a minimum of 7 inches measured vertically between the lowest point on the steering wheel rim and a line parallel to the floor and passing through the highest point of the unoccupied seat cushion when steering wheel is in the operating position. There shall be a minimum of 3 inches between the rim of the steering wheel and the directional control lever(s) for all positions of the lever(s). The steering wheel shall be capable of withstanding a minimum 300 pound force applied perpendicular to the plane of the steering wheel, and a minimum 150 pound force applied tangentially to the rim of the steering wheel, without permanent deformation or damage. The steering system shall be capable of being turned from stop to stop at low engine idle (for hydraulic-mechanical drive) in not more than 5 seconds and not less than 3 seconds at governed engine speed.

3.17 Brake systems. The truck shall be equipped with a service brake or regenerative brake system, parking brake system and emergency brake system capable of safely stopping, holding and controlling the movement of the truck

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on the slopes specified herein and under the operating conditions and in the environments specified herein with and without rated load.

3.17.1 Brake systems: Hydraulic mechanical drive.

3.17.1.1 Service brake system performance. The service braking system shall provide braking for all wheels. When operating in the forward and rearward direction, the system shall be capable of bringing the truck, with rated load in the carry position, to a complete stop without any wheel leaving the ground and with fishtailing of no more than 1 foot from a speed of not less than 9 mph with a pedal force not to exceed 90 pounds as follows:

(a) Dry - within 21 feet when tested as specified in 4.6.2.18.
(b) Fade - within 21 feet when tested as specified in 4.6.2.18.
(c) Wet - within 21 feet when tested as specified in 4.6.2.18.

The truck shall be capable of a panic stop from maximum truck speed in the reverse direction with rated load in the carry position with fishtailing of no more than 1 foot and without any wheel leaving the ground when tested as specified in 4.6.2.18. The truck without load shall be capable of a panic stop from maximum speed in the reverse direction and from maximum speed in the forward direction with fishtailing of not more than 1 foot and without any wheel leaving the ground when tested as specified in 4.6.2.18. Panic stop is defined as a pedal application of greater than 150 pounds.

3.17.1.2 Power brake system. The power source shall charge the system to maximum power setting within 2 minutes when the brake actuating energy is fully exhausted. With no further addition of brake system actuation energy, a sufficient energy reserve shall be furnished such that a fully charged system shall have the capacity of six full effective service brake applications. An audible and visual low pressure warning device shall be furnished.

3.17.1.2.1 Straight airbrake system. When a straight airbrake system is furnished, the system shall include not less than a 12 cfm air compressor, alcohol evaporator, and wet and dry reservoirs. The air compressor shall be engine driven, engine lubricated and water cooled. All reservoirs shall shall be furnished with automatic or remote control drain valves. All lines (except drains) connected to the reservoirs shall be attached on or above the horizontal centerline of the reservoirs. The airbrake plumbing shall be routed, supported and fastened to provide maximum protection. In-cab plumbing and chassis plumbing not subject to road hazards shall conform to SAE J844, Type 3. Chassis plumbing subject to road hazards shall conform to SAE J1402, Type D, Class II, flexible hose with SAE 100R5 reusable fittings as specified in SAE J516. The air compressor discharge line shall be teflon or teflon with copper tubing.

3.17.1.2.2 Air-over-hydraulic brake system. When an air-over-hydraulic brake system is furnished, the complete air system shall be in accordance with 3.17.1.2.1. Hydraulic brake hose shall conform to SAE J1401.

3.17.1.3 Parking brake(s). Parking brake(s) shall hold, but not necessarily stop the truck without rated load when the truck is stopped on a 25 percent grade in both the ascending and descending position. The parking

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brakes shall be operated by controls in accordance with SAE J680 or shall be hand lever operated. The hand lever shall have a locking device. The manual control shall be located within reach of the operator and in a non-interfering position. Hand lever pull to set the parking brake shall not exceed 90 pounds. The parking brake shall be an independent friction brake mounted on the drive or propeller shaft rotating with the wheels, or a separate isolated system actuating the service brakes. Adjustment of parking brake shoes or bands shall be accomplished without the removal of any assembly.

3.17.1.4 Emergency braking provisions. The truck shall provide not less than six "dead engine" stops with rated load while descending the 25 percent grade. The emergency braking provisions may be provided through either the parking brake or service brake systems.

3.17.2 Brake system: Electric drive. The truck shall be furnished with a regenerative brake system, parking brake system and an emergency brake system.

3.17.2.1 Regenerative, brake system. The regenerative brake system shall provide braking for all wheels. When operating in the forward and rearward direction, the regenerative system shall be capable of bringing the truck with rated load in carry position to a complete stop without any wheel leaving the ground and with fishtailing of no more than 1 foot from a speed of not less than 9 mph as follows:

(a) Dry - within 32 feet when tested as specified in 4.6.2.18.
(b) Fade - within 32 feet when tested as specified in 4.6.2.18.
(c) Wet - within 32 feet when tested as specified in 4.6.2.18.

3.17.2.2 Parking brake system. The parking brake(s) shall hold, but not necessarily stop the truck without rated load when the truck is stopped on a 25 percent grade in both the ascending and descending position.

3.17.2.3 Emergency brake system. The combination of the regenerative system and the emergency brake system shall be capable of bringing the truck with rated load in carry position to a complete stop from a speed of not less than 9 mph in reverse within 21 feet when tested as specified in 4.6.2.18. Test Method No. 15 only shall be required for the combination brake system test.

3.18 Operator access. Operator access systems to the truck shall be in accordance with minimum requirements of SAE J185.

3.19 Seat. A cushion seat and backrest shall be furnished in accordance with the minimum requirements specified in SAE J899. Horizontal seat adjustment shall be by the operator from the operating position without the use of tools. The seat cushion and backrest shall be foam rubber or polyurethane, covered with slip coated vinyl upholstery having a minimum weight of 40 ounces per yard for 54 inch width. The seat location shall conform to SAE J898.

3.20 Hood. The engine shall be protected by a hood cover and sidepanels, except that sidepanels may be omitted when wheal guards provide the engine compartment with protection from road splash and debris. Sidepanels shall be

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quick opening if removal of the side panel is required for operator servicing of the truck. The service sidepanels shall be capable of being opened or removed by one man and shall be within the design weight limits of MIL-STD-1472. Handtools shall not be required for release of quick opening sidepanels. Elastomer hood latch or paddle latch shall be provided on the hood cover and sidepanels. All engine accessories shall be completely accessible, without requiring the removal of any additional truck structure, when the service sidepanels are opened.

3.21 Instrument Panel. The instrument panel shall be located so that the operator can read the instruments from his normal operating position. The instrument panel shall be installed on an inclined plane. On this panel shall be mounted the electric system monitor, engine oil pressure monitor, engine coolant temperature monitor, torque converter or transmission fluid temperature monitor, fuel gage (or high and low indicator lights), engine run control, light switches, starter switch, airbrake pressure gage, if applicable, and any other instruments used, unless otherwise specified herein. All gages and switches shall be identified on the face of the gage or as specified in 3.31.1 as to nomenclature of component and fluid being monitored and positions and function control, (international symbols are acceptable). All gages shall be flush mounted. The instrument panel shall be provided with switch controlled illumination for other than daylight visibility.

3.22 Controls and instrumentation. Each control shall be provided with position markings either on plates conforming to 3.31.1 or embossed or depressed on or near the control. Markings for levers which control any part of the lift mechanism shall conform to ANSI MH11.3. The markings shall be visible to the operator from his normal operating position. Letter height for control markings shall be equal to or greater than specified for general dial and panel design for the applicable viewing distance of the 95th percentile man, with seat adjusted to rear position, as specified in MIL-STD-1472. All controls for the truck shall be located in the operator's compartment and shall be within the maximum areas specified in SAE J898. Unless otherwise specified herein, the distance from the near edge of one control to the near edge of an adjacent control or other surface shall be a minimum of 1-1/2 inches. Electric controls and instrumentation shall be moisture and weather resistant. Mechanical push-pull remote controls, when furnished, shall be of the waterproof antifriction type, conforming to requirements for marine cables specified in SAE J917, except as specified herein. As a minimum, all exposed parts of the control cables shall be fabricated of corrosion resistant steel. The cables shall be covered with a polyethylene or vinyl jacket. Cable ends shall be tailored to actuate all levers through their respective arcs without binding. Bend radii shall be as recommended by the cable manufacturer. The "as installed" mechanical efficiency of each installation shall be not less than 70 percent, and the no load friction shall not exceed 5 ounces per foot. The manufacturer's published ratings for continuous operation shall not be exceeded for any required operation of the control specified herein. When a cab is furnished, only controls associated with the cab, such as a windshield wiper and heater controls, may be mounted on the cab structure.

3.22.1 Light switches. A minimum of three light switches shall be furnished located within reach of the operator. One switch shall control the forward floodlight(s) as specified in 3.25.1 and taillight as specified in

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3.25.2. One switch shall control the rear floodlight(s) as specified in 3.25.1. One switch shall control the twistlock lights specified in 3.25.1.

3.22.2 Engine run control. An engine run control shall be furnished and shall be a key type. It shall have not less than two positions "on" and "off". The switch shall control the operation of monitors and of all electrical equipment except the horn. A mechanical engine shutoff may be furnished and shall be in addition to the emergency shutoff as specified (see 3.10). If mechanical engine shutoff and electrical fuel pump are furnished a warning buzzer shall be furnished to sound when key is "on" and engine is not running. Two engine run control keys shall be furnished with each truck. All engine run control keys shall be identical for every truck.

3.22.3 Starting system control. The starting system solenoids (see 3.10.2.3), shall be operated by a pushbutton starter switch or a third position on the engine run control switch. The starter shall be inoperable when the engine run control is in the "off" position.

3.22.4 Braking and declutch pedals. Two pedals or treadles shall be provided for declutch and braking control for trucks furnished with hydraulic mechanical drive train. The right pedal or treadle located for convenient right foot operation shall provide braking control only. The left pedal or treadle located for convenient left foot operation shall be a combination declutch and braking control. Each control shall be a minimum of 2-1/2 inches wide and shall have an all around clearance of not less than 1-1/2 inches. Grease fittings or permanent lubrication shall be provided for lubricating moving parts on all pedals and shafts. For the electric drive trucks the service brake system may be activated by returning the speed controller to zero speed position.

3.22.5 Speed control.

3.22.5.1 Hydraulic mechanical drive. The speed control shall be installed for comfortable right foot operation and shall be located to the right of, and shall be free of interference from the brake pedal. The speed control shall have an all around clearance of not less than 3 inches. The speed control shall be furnished with a spring return to engine idle. The actuation force or the speed control measured at the top shall be a minimum of 10 pounds and a maximum of 20 pounds through the total pedal travel with engine running.

3.22.5.2 Electric drive. The speed control shall be hand operated and located on the instrument panel.

3.22.6 Directional controls.

3.22.6.1 Hydraulic mechanical drive transmission controls. Selective forward and reverse and speed range controls shall be provided and shall be located for convenient left hand operation. Movement of the control lever in a forward or upward direction shall cause forward motion of the truck. Transmission control levers and linkage shall withstand, without permanent deformation or damage, a force of 100 pounds at the handgrip and the shock and vibration encountered during operation specified herein.

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3.22.6.2 Electric drive direction controls. Forward and reverse direction controls shall be provided and shall be combined with the speed controller for convenient right hand operation.

3.22.7 Container handling controls. Lift, tilt, oscillation (if powered), sideshift, and twist lock controls shall be self-centering levers or toggle switches, located for convenient right hand operation within the maximum areas specified in SAE J898. All control levers and linkage shall withstand) without permanent deformation or damage, a force of 100 pounds at the handgrip and the shock and vibration encountered during operations specified herein. The force necessary to move any control to the end of its travel shall not exceed 20 pounds at the handgrip. Total control lever travel from one end to the other shall not exceed 14 inches, measured at the center of the handgrip. When separate lift and tilt levers are used, the tilt control shall be at the right of or below the lift control, as viewed by the operator. Control functions and predominant control motion shall be as follows (function listed, followed by control motion - the first applicable motion listed, depending on lever mounting, shall apply):

(a) Lift load - Toward the operator, rearward, or upward (opposite direction to lower).
(b) Tilt load rearward - Toward the operator, rearwards or in the same rotational direction as load movement (opposite direction to tilt forward).
(c) Sideshift to right - To the operator's right, rearward, or upward (opposite direction to sideshift left).
(d) Oscillate clockwise (as viewed by operator) - To the operator's right, rearward, or upward (opposite direction to oscillate counterclockwise).

3.22.8 Electrical system monitor. A remote indicating ammeter or voltmeter, or alternator indicator red light shall be furnished. The ammeter shall be of sufficient capacity to indicate variations in current supply or demand experienced by the truck, exclusive of starter requirements. The indicator light shall light when the alternator is not charging and the engine run control is "on".

3.22.9 Engine oil pressure monitors. A remote indicating engine oil pressure gage shall be furnished. A low pressure red indicator light or low pressure alarm shall be furnished in addition to the gage. Means shall be provided to check operation of the light or alarm.

3.22.10 Engine coolant temperature monitor. A remote indicating engine coolant temperature gage or a high temperature red indicator light or a high temperature alarm shall be furnished. If an indicator light or alarm is furnished, means shall be provided to check operation of the light or alarm.

3.22.11 Transmission temperature monitor for hydraulic mechanical drive trucks. A remote indicating torque converter or transmission temperature gage or a high temperature red indicator light shall be furnished. If an indicator light is furnished, means shall be provided to check operation of the light.

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3.22.12 Hourmeter. An operational hourmeter which registers the number of engine operating hours shall be provided. The hourmeter shall be accessible for reading without moving any component of the truck.

3.22.13 Fuel monitor. An electric, remote indicating fuel gage or fuel tank dipstick shall be provided.

3.22.14 Air pressure monitor. When a compressed air system is furnished, an air pressure gage shall be furnished and an audible alarm shall indicate when air pressure is inadequate for proper brake operations.

3.23 Battery and battery mounting.

3.23.1 Batteries. A minimum of four batteries conforming to SAE J530, Group 8D, Type 21T2 or conforming to SAE J537, Type 20T8A shall be furnished. The batteries shall be arranged for 24 volts. The batteries shall be shipped charged and dry. Electrolyte shall be furnished in a separate container as specified in MIL-STD-162. The batteries shall be negative grounded in accordance with SAE J538.

3.23.2 Battery mounting. Battery mounting shall be in a weathertight compartment having a no-sag cover. The compartment shall be accessible for battery removal without chafing or other damage to the battery. Batteries shall be located so they can be cleaned, serviced, and removed without removing any component except the quick-release battery box cover if a battery box is furnished. Battery supports, holddowns, and areas around the installation which could possibly by affected by dripping or seepage of acids shall be protected with a coating conforming to TT-C-494, Type II. The battery shall be mounted in such a manner as not to interfere with access to engine components (accessories). Battery mounting shall provide for complete support over the entire base of the battery and shall be in such a position that the level of the electrolyte is directly visible without removing the battery from its mounting bracket or requiring the use of tools. Battery restraining clamps shall not be designed for or dependent upon special battery shapes and shall be provided to hold the battery in a fixed position. The battery compartment shall have provision for drainage, and shall have provisions for gas venting at or near the top of the compartment. No-sag cover and positioning shall be protected against short circuiting. The cables shall be protected by rubber grommets or insulated passages at entry to the battery box.

3.23.3 Battery cable terminals. Battery cables shall be in accordance with SAE J541. Positive and negative cable terminals shall be identified. Terminals secured with a plated corrosion resistant bolt and nut shall be provided on the ends of cables connected to the battery to facilitate disconnection and battery removal.

3.23.4 Slaving components. The truck shall be equipped with a 24 volt slave receptacle conforming to MS52131. The slave receptacle shall permit charging of the truck batteries and slave starting of the engine from an external power source and shall also provide a power source for charging and slaving other equipment. The slave receptacle shall be installed on the exterior of the truck near the battery enclosure and shall be accessible to

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personnel standing on the ground. The voltage potential of the slave receptacle shall be identified on a plate furnished adjacent to the receptacle. The plats shall be in accordance with 3.31.1.

3.24 Pintle hook and towing lugs.

3.24.1 Pintle hook. A rear mounted pintle hook conforming to MS51117 shall be provided. The pintle hook shall be located on the vertical centerline of the truck. The pintle hook mounting shall be of sufficient strength to withstand, without permanent deformation or damage, the drawbar pull induced when the pintle hook is restraining forward motion at torque converter stall or electric drive line stall.

3.24.2 Towing lugs. Two vehicular towing lugs, conforming to MS500004, for tanks (above 50 tons), shall be furnished and shall provide clearance to connect a towbar conforming to MS500048, heavy duty capacity. The lugs shall be installed between 17 to 21 inches on each side of the truck centerline, on the rear of the truck.

3.25 Electrical system. Multiple conductor routing shall be protected by flexible tubing, plastic electrical insulation tape, or woven vinyl coated nylon braiding yarn. Grommets shall be provided whenever wiring passes through bulkheads, partitions, or structural members; and the wiring shall be fastened to prevent chafing or abrasion. All wiring shall be color or number coded in accordance with SAE J821. All 24 volt circuits except starting shall be furnished with circuit breakers in accordance with SAE J553. High voltage and high amp circuits on electric drive trucks may be fused. All electrical connections fastened with terminal nuts shall be equipped with lock washers or nut locking devices. Circuit breakers shall be grouped in convenient locations and suitably marked for size and function.

3.25.1 Lights. Each truck shall be equipped with a minimum of six sealed beam CIM floodlamps or six 500 watt minimum quartz floodlamps. The CIM floodlamps sealed beam units shall conform to SAE J598t PAR 46, Trade No. 4578 which shall be shock mounted in elastomer ring housings. Each floodlight shall be adjustable in accordance with SAE J598. Lights shall be arranged to provide forward illumination for driving when the truck is carrying a container at all lift heights from ground level to maximum lift height. Two lights shall be adjustable to illuminate to the front of the truck. Two lights shall be adjustable to illuminate to the rear of the truck. Two lights shall illuminate the two near twistlocks for all lift heights or shall be adjustable from the operator's position. Adequate protection against damage of each light, by means of position or guard, shall be provided. If floodlamps are mounted on the ROPS they shall be provided with stud and terminal connectors near the place of attachment of the ROPS to the truck chassis. All floodlamps shall be adjustable in accordance with SAE J598. All lights shall be positioned within the outline of the truck.

3.25.2 Taillights. Two taillight-stoplight assemblies in accordance with SAE J585 and SAE J586 shall be installed, one on the left rear and one on the right rear of the truck, with the lens face recessed not less than 1/2 inch back of a protecting member.

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3.25.3 Horn. The truck shall be equipped with an electric, air, or airover-electric horn. The horn button assembly and electrical wiring for the horn shall be constructed to be moisture and weather resistant to prevent entry of moisture when operated or stored outdoors under all weather conditions. The horn shall be operable from the operator's seat.

3.26 Personnel cab. The truck shall be equipped with an insulated, weathertight personnel cab, which may be combined with ROPS and FOPS specified in 3.5.1. The cab clearances shall be in accordance with SAE J154. The cab shall withstand all operational shock and vibration without deformation or damage under the operating conditions specified herein. The cab shall be constructed of metal and shall be equipped with a minimum of one door and another emergency exit (either door or window that can be opened) on the other side of the cab. All door's shall have nonlocking closing latches, opening stops and hold-open catches. The doors shall either slide and latch at each end of travel or shall swing open a minimum of 90 degrees and latch open. All hardware shall be of the heavy duty truck or industrial type resistant to corrosion. Front and rear window defrosters, operating in conjunction with the heater blower (see 3.26.4), or not less than two ventilation fans, not less than 200 cfm inclosed in a wire guard, shall be furnished. Fans shall be directionally adjustable to direct air flow toward all glazing. As a minimum, each fan shall have an "on-off" switch.

3.26.1 Glazing. As a minimum, glazing shall include a windshield, door windows and a rear window. Other windows, as necessary, shall be provided to give an upright seated operator all around visibility and visibility of the twistlocks in all possible positions. As a minimum, two windows, one each on opposite sides of the cab, shall be capable of being opened. When a roof window is provided, it shall be sloped to provide for water drainage and guarded as necessary to meet the FOPS requirements (see 3.5.1). Glazing shall conform to ANSI Z26.1, Type AS1 for the windshield and roof window and Type AS1 or AS2 for other windows. Fixed or hinged glazing shall be sealed in waterproof, stress absorbent molding and movable windows shall be sealed on extra heavy type, pile lined channels.

3.26.2 Interior. The interior of the cab shall be fitted with insulation, retarding both heat lose and noise intrusion. Heavy-duty floor mats shall be furnished. All cab openings shall be sealed with elastomer grommets, seals, weather stripping, or body compound.

3.26.3 Windshield and window wipers. The cab shall be equipped with at least two powered windshield wiper assemblies to clear the windshield and the rear window. The wiper motors shall provide not less than 90 inch-pounds of torque when electric motors are furnished, or provide not less than 170 inch-pounds of torque when air motors are furnished. Wipers shall be capable of variable stroke speeds up to at least 60 strokes per minute. Wiper arms and blades shall be of the heavy duty commercial type, having saddle-and-pin blade fastening. The wiper arms shall provide tension on the glazed surface of at least 1 ounce for each inch of wiper arm length. Air motors shall be plumbed to exhaust air outside of the cab. The operator controls shall be either installed on instrument panels or within easy reach of the operator within the cab.

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3.26.4 Heater. Recirculating air type heater(s) shall be furnished. The heater(s) shall have a total rating of at least 40,000 Btu/hour. The heater(s) controls shall be within reach of the operator and shall provide for variable selection of blower speeds.

3.27 Electromagnetic interference. The electromagnetic interference emission characteristics of the truck shall meet the limits specified in MIL-STD-461, Class IIIC, except that a 20 dB relaxation is permitted in the emission limits.

3.28 Lubrication. Unless otherwise specified herein, the truck shall be serviced with military lubricants selected by the contractor (see 6.7). Run-in and testing shall be performed with these lubricants as applicable for the ambient temperature.

3.28.1 Lubrication fittings. Lubrication fittings shall conform to SAE J534. Fittings shall be located in a protected position. Fittings shall be accessible to a minimum 15 ounce grease gun with a 10 inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand operated fasteners may be removed to provide accessibility.

3.28.2 Enclosures with integral reservoir of lubricant. Enclosures such as the gearcases, transmission housing and engine crankcase, which contain a reservoir of liquid lubricants for the lubrication of the parts enclosed, shall be equipped with dipsticks, check plugs not less than 1/2-inch pipe size, or sight gages to determine the acceptable level of the lubricant. As a minimum, dipsticks shall be graduated to show full and add levels. Each enclosure shall be equipped with a means for filling the enclosure with lubricant and for draining. The drains, except planetaries, shall be fitted with a magnetic drainplug having either tapered pipe threads or shoulder and gasket. The drainplug shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage, except from planetary hubs, shall be to the ground when the truck is on level ground. Integral tubes or troughs may be used to convey the lubricant from the drain to the ground. Accessibility to the drainplug, the filling means, and the lubricant level checking device shall be provided without the removal or adjustment of accessories or parts. Plates equipped with hand operated fasteners may be removed.

3.29 Fungus and moisture resistance. Electrical circuitry, including all components and connections except as specified below, shall be protected from the effect of fungus growth and moisture by an overall treatment with with a varnish conforming to MIL-V-173, composition specified in 3.29.1, with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish, or a varnish capable of providing the same protection and having the same or equal antipollution ingredients when required:

(a) Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

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When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter that would interfere with the adherence or function of the varnish.

3.29.1 Composition I. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of Composition I. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 102, Southern California Air Pollution Control District.

3.30 Slinging and tiedown provisions. Permanently affixed slinging and tiedown provisions that enable the truck to be lifted completely assembled and in its normal operational mode, and to be fastened to the floor or deck of a transportation medium, shall be provided. Provisions shall conform to MIL-STD-209, Classes 1 and 2 or Class 3 for Type II equipment. A steering frame lock in accordance with SAE J276 shall be provided. The frame lock shall be painted in accordance with 3.32.

3.31 Identification marking. Each truck shall be identified in accordance with MIL-STD-130 on identification plates conforming to MIL-P-514, Type I, Style 1, Composition C (GG-P-455, Type I, Grade A, Class 1) material. Plates shall be furnished and mounted by the contractor. Shipping data plates for each highway transport load (see 3.1.2.1) shall be furnished on the truck chassis. The plates shall conform to MIL-P-514, Type III, Composition C (GG-P-455, Type I, Grade A, Class 1) material and, in addition, shall indicate the silhouette of each transport load in transport position indicating the center of gravity, the location and capacity of the slinging and the tiedown provisions. The plates shall be attached by screws, drive screws, bolts, or rivets in conspicuous protected locations.

3.31.1 Instruction plates. The truck shall be equipped with instruction plates conforming to MIL-P-514, Type III, Composition C (GG-P-455, Type I, Grade A, Class 1) material or commercially proven decals, including warnings, cautions, and diagrams describing any special or important procedures to be followed in assembling, operating, towing, or servicing the truck.

3.32 Treatment and painting. The portions of the truck normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A, followed by a top coat of MIL-E-52798 or MIL-E-52835 alkyd enamel, forest green.

3.33 Marking. Registration and other markings shall be applied in accordance with MIL-STD-642, color black. The gross weight of the truck shall be marked on each side of the truck. The prescribed tire pressure shall be marked in a suitable location adjacent to each tire with block or stencil type letters not more than 1 inch and not less than 3/4 inch high. All suitable slinging and tiedown provisions shall be marked.

3.34 Workmanship.

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3.34.1 Castings and forgings. All parts, components, and assemblies of the truck which includes castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

3.34.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly, except in the performance of designed heat treatment. External surfaces shall be free from burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.34.3 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the truck to conform to the examination and test requirements specified in Section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible, and in no event, shall be separated by more than 3/16 inch, unless appropriate bridging techniques are used. The welding process used in fabrication of the truck shall be at the option of the contractor.

3.34.3.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

Structural Welding Code, AWS D1.1, Qualification, Section 5 Welding Qualifications of the ASME, Section IX.

Contractors who only make horizontal welds need not qualify welders for "all position welding." Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in in accordance with AWS and ASME codes.

3.34.3.2 Weld-quality.

(a) The contractor shall be responsible for determining the inspections needed to insure that, when the welded parts are

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assembled together to make the truck, the truck shall conform to the inspection requirements specified in Section 4.

(b) All weldments shall be free of slag, flux, weld, spatter, and other impurities detrimental to either the appearance or strength of the weldment.

(c) Undercut in weldments shall be held to a minimum and shall not be more than 1/32 inch deep.

3.34.3.3 Welding-practices.

(a) Preheat of materials being welded and maximum interpass temperature during welding shall be in accordance with the contractor's recommendations.

(b) Tack welds shall be subject to the same quality requirements as final welds.

(c) Work shall be positioned for flat welding whenever practicable.

(d) Procedures and sequences shall be such that distortion and shrinkage will be held to a minimum. When straightening is required, caution shall be exercised to insure that the straightening process does not weaken the part and truck.

3.34.4 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers, lockwashers, or other positive locking devices shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capserews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.34.5 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492 or IFI Standard No. 114 for Break Mandrel Blind Rivets.

3.34.6 Machine work. Tolerances and gages for metal fits shall conform to the limits specified herein and to the standards of the Materials Handling Equipment Industry or the Construction Equipment Industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with the requirements of referenced specifications and standards.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection. Prior to examination and test of the truck specified in 4.3.1 and 4.3.2, the following shall be performed:

(a) Service the truck with the oils, greases, and fuel specified herein and designated for use in the ambient temperatures at which the tests will be conducted.
(b) Conduct the run-in as prescribed by the contractor (see 4.6.2.37).
(c) Set the hydraulic systems relief pressure as specified by the contractor and in 3.12.1 and record. Any change made to this setting shall constitute failure of all tests.
(d) Furnish a schedule of maintenance to be followed during all testing of the trucks. A list of tools required to perform this maintenance shall also be provided.

4.3.1 Preproduction examination.

4.3.1.1 Pretest examination. Prior to testing, one truck shall be examined for the defects marked "X" in Column 1 of Table II. Presence of one or more defects shall be cause for rejection.

4.3.1.2 Post-test disassembly, and examination. Upon successful completion of all tests specified in 4.3.2, the truck shall be subjected to the tests marked "X" in Column 3 of Table III and 4.6.2.38. Presence of one or more defects shall be cause for rejection.

4.3.2 Preproduction tests. Upon successful completion of the examination specified in 4.3.1.1, one preproduction truck shall be subjected to the tests marked "X" in Column 1 of Table III except for the endurance tests (see 4.6.2.34). The other preproduction truck shall be subjected to the endurance tests (see 4.6.2.34). Tests may be performed in any order. Except as otherwise specified herein, test conditions shall apply throughout testing. Failure of any test shall be cause for rejection. Failure of a preproduction truck to meet any requirement specified herein shall be cause for rejection of the preproduction trucks.

4.4 Initial production inspection.

4.4.1 Initial production examination and tests. When specified (see 3.2.2), two or more initial production trucks will be selected at random by the Government from the trucks being produced by production tooling and will be examined for the defects marked "X" in Column 2 of Table II and subjected to the tests marked "X" in Column 2 of Table III to determine conformance to the requirements of this specification. The inspection will be performed by

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the Government at a site selected by the Government. Acceptance of an Initial production truck shall not exclude the remianing trucks from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1.1 Inspection failure. Failure of an initial production truck to meet any requirement herein during and as a result of the examination and tests specified in 4.4.1 shall be cause for rejection of the initial production truck(s), and shall be cause for refusal by the Government to continue acceptance of production trucks until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on trucks previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all trucks accepted prior to the completion of initial production inspection are similarly deficient, unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each truck shall be examined for the defects marked "X" in Column 3 of Table II. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each truck shall be tested as specified in Column 4 of Table III. Failure of any test shall be cause for rejection.

4.6 Inspection schedule.

4.6.1 Examination. Trucks shall be examined as specified herein in accordance with Table II.

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Table II. Examination Schedule

Preproduction	Initial Production	Quality Conformance	Defect	Requirement Paragraph
1	2	3	4	5
X	X	-	101. Dissimilar metals not insulated as specified.	3.3.1
X	X	-	102. Screw threads not as specified.	3.3.2
X	X	X	103. Tire material not as specified.	3.3.3
X	X	X	104. Safety provisions not as specified.	3.5
X	X	X	105. ROPS and FOPS missing or not as specified.	3.5.1
X	X	X	106. Seatbelt missing or not as specified.	3.5.2
X	X	X	107. Backup alarm missing or not as specified.	3.5.3
X	X	-	108. Human factors not as specified.	3.6
X	X	-	109. Maintainability provisions not as specified.	3.7
X	X	-	110. Calculated flotation index not as specified.	3.9.2
X	X	-	111. Engine not as specified.	3.10
X	X	X	112. Starting priming system missing or not as specified.	3.10.2.1
X	X	-	113. Alternator not as specified.	3.10.2.2
X	X	-	114. Starting system not as specified.	3.10.2.3
X	X	-	115. Fuel system not as specified.	3.10.2.4
X	X	-	116. Air cleaner not as specified.	3.10.2.5
X	X	-	117. Cooling system not as specified.	3.10.2.6
X	X	-	118. Fan not as specified.	3.10.2.6.1
X	X	-	119. Radiator not as specified.	3.10.2.6.2
X	X	-	120. Engine coolant hose not as specified.	3.10.2.6.3
X	X	-	121. Exhaust system not as specified.	3.10.2.8
X	X	-	122. Crankcase ventilation system not as specified.	3.10.2.9
X	X	-	123. Oil filters not as specified.	3.10.2.10
X	X	-	124. Drive train assembly not as specified.	3.11
X	X	-	125. Hydraulic mechanical drive train not as specified.	3.11.1
X	X	-	126. Powershift transmission not as specified.	3.11.1.1
X	X	-	127. Electric drive train not as specified.	3.11.2
X	X	-	128. DC generator not as specified.	3.11.2.1
X	X	-	129. Switchboard not as specified.	3.11.2.2
X	X	-	130. DC motors not as specified.	3.11.2.3

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Table II. Examination Schedule (Cont'd)

1	2	3	4	5
X	X	-	131. Hydraulic system not as specified.	3.12.1
X	X	-	132. Hydraulic lines not as specified.	3.12.1.1
X	X	-	133. Pressure hose assemblies not as specified.	3.12.1.1.1
X	X	-	134. Pressure tubing not as specified.	3.12.1.1.2
X	X	-	135. Pressure tube fittings not as specified.	3.12.1.1.3
X	X	-	136. Suction line not as specified.	3.12.1.1.4
X	X	-	137. Return lines not as specified.	3.12.1.1.5
X	X	-	138. Quick disconnects not as specified.	3.12.1.1.6
X	X	-	139. System filtration not as specified.	3.12.1.3 thru 3.12.1.3.3
X	X	-	140. Reservoir not as specified.	3.12.1.4
X	X	-	141. Relief valve not as specified.	3.12.1.5
X	X	-	142. Hydraulic control valves not as specified.	3.12.1.6
X	X	-	143. Cylinders not as specified.	3.12.1.7
X	X	-	144. Hose reels not as specified.	3.12.1.8
X	X	-	145. Electric power system not as specified.	3.12.2
X	X	-	146. AC generator not as specified.	3.12.2.1
X	X	-	147. AC motors not as specified.	3.12.2.2
X	X	-	148. Mast and carriage not as specified.	3.13
X	X	-	149. Container tophandlers not as specified.	3.13.1
X	X	-	150. Coupling devices not as specified.	3.13.1.1
X	X	X	151. Alining arms not as specified.	3.13.1.2
X	X	X	152. Coupler indicator lights not as specified.	3.13.1.3
X	X	-	153. Fork kit not as specified.	3.13.2
X	-	-	154. Forks not as specified.	3.13.2.1
X	X	-	155. Wheels not as specified.	3.15.1
X	X	-	156. Tires not as specified.	3.15.2
X	-	-	157. Steering control not as specified.	3.16.1
X	X	-	158. Brake system: hydraulic mechanical drive not as specified.	3.17.1
X	X	-	159. Service brake system not as specified.	3.17.1.1
X	X	-	160. Power brake system not as specified.	3.17.1.2
X	X	-	161. Straight air brake system not as specified.	3.17.1.2.1
X	X	-	162. Air over hydraulic brake system not as specified.	3.17.1.2.2

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Table II. Examination Schedule (Cont'd)

1	2	3	4	5
X	X	-	163. Parking brakes not as specified.	3.17.1.3 3.17.2.2
X	X	-	164. Brake system: electric drive not as specified.	3.17.2 3.17.2.1
X	X	-	165. Regenerative brake system not as specified.	3.18
X	X	-	166. Operator access not as specified.	3.19
X	X	-	167. Seat not as specified.	3.20
X	X	-	168. Hood not as specified.	3.21
X	X	-	169. Instrument panel not as specified.	3.22
X	X	-	170. Controls and instrumentation not as specified.	3.22.1
X	X	-	171. Light switches missing or not as specified.	3.22.2
X	X	-	172. Engine run control not as specified.	3.22.3
X	X	-	173. Starting system control not as specified.	3.22.4
X	X	-	174. Braking and declutch controls not as specified.	3.22.5.1,
X	X	-	175. Speed control not as specified.	3.22.5.2 3.22.6.1
X	X	-	176. Hydraulic mechanical drive transmission controls not as specified.	3.22.6.2
X	X	-	177. Electric drive direction controls not as specified.	3.22.7
X	X	-	178. Container handler controls not as specified.	3.22.8
X	X	-	179. Electrical system monitor missing or not as specified.	3.22.9
X	X	-	180. Engine oil pressure monitors missing or not as specified.	3.22.10
X	X	-	181. Engine coolant temperature monitor missing or not as specified.	3.22.11
X	X	-	182. Transmission temperature monitor for hydraulic mechanical drive trucks missing or not as specified.	3.22.12
X	X	-	183. Hourmeter missing or not as specified.	3.22.13
X	X	-	184. Fuel monitor missing or not as specified.	3.22.14
X	X	-	185. Air pressure monitor missing or not as specified.	3.23.1
X	X	X	186. Batteries not as specified.	3.23.2
X	X	-	187. Battery mounting not as specified.	3.23.3
X	X	-	188. Battery cable terminals not as specified.	3.23.4
X	X	-	189. Slave receptacles missing or not as specified.	

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Table II. Examination Schedule (Cont'd)

1	2	3	4	5
X	X	-	190. Pintle hook missing or not as specified.	3.24.1
X	X	-	191. Towing lugs missing or not as specified.	3.24.2
X	X	-	192. Electrical system not as specified.	3.25
X	X	X	193. Lights missing or not as specified.	3.25.1
X	X	X	194. Taillights missing or not as specified.	3.25.2
X	X	X	195. Horn missing or not as specified.	3.25.3
X	X	-	196. Personnel cab not as specified.	3.26
X	X	-	197. Glazing not as specified.	3.26.1
X	X	-	198. Cab interior not as specified.	3.26.2
X	X	-	199. Windshield and window wipers not as specified.	3.26.3
X	X	-	201. Heater missing or not as specified.	3.26.4
X	X	-	202. Lubrication not as specified.	3.28
X	X	-	203. Lubrication fittings not as specified.	3.28.1
X	X	-	204. Filling, draining, and checking provisions for lubricant enclosures missing or not as specified.	3.28.2
X	X	-	205. Fungus and moisture resistance provisions not as specified.	3.29, 3.29.1
X	X	X	206. Slinging and tiedown provisions and frame lock missing or not as specified.	3.30
X	X	X	207. Identification marking missing or not as specified.	3.31
X	X	-	208. Instruction plates missing or not as specified.	3.31.1
X	X	-	209. Treatment and painting not as specified.	3.32
X	X	X	210. Marking not as specified.	3.33
X	X	X	211. Castings and forgings not as specified.	3.34.1
X	X	X	212. Metal fabrication not as specified.	3.34.2
X	X	X	213. Welding not as specified.	3.34.3 thru 3.34.3.3
X	X	X	214. Bolted connections not as specified.	3.34.4
X	X	X	215. Riveted connections not as specified.	3.34.5
X	X	X	216. Machine work not as specified.	3.34.6
X	X	X	217. Parts or components missing or do not function.	Section 3
X	X	X	218. Assembly incorrect or not complete.	Section 3

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4.6.2 Tests. Tests shall be conducted in accordance with Table III. An "X" in the column indicates the tests that shall be conducted.

4.6.2.1 Test conditions. Unless otherwise specified herein, tests shall be conducted at the ambient temperature and climatic conditions existing at the place of test. Only that maintenance established by the contractor and submitted as maintenance schedule prior to commencement of the tests shall be performed during the tests. Safety and human factors considerations (see 3.5 and 3.6), shall be evaluated throughout testing. Test methods referenced in tests specified herein are those contained in the Appendix. Operation of the truck shall be with fuels, oils, lubricants, and antifreeze specified herein as applicable for the temperature conditions existing at place of test. All tests shall be conducted at the contractor's recommended tire pressure.

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Table III. Test Schedule

Preproduction	Initial Production	Posttest	Quality Conformance	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6	7
X	X	X	X	Test conditions.	4.6.2.1	3.5, 3.6, 3.8
X	X	-	X	Engine.	4.6.2.2	3.10
X	X	-	-	Lift height, low engagement weight and dimensions.	4.6.2.3	3.9.8, 3.9.9
X	X	X	-	Lifting and lowering speed.	4.6.2.4	3.9.6, 3.9.7
X	X	-	-	Curb clearance circle.	4.6.2.5	3.9.18
X	X	X	-	Travel speed.	4.6.2.6	3.9.4, 3.10.2.8, 3.11.1.1
X	X	-	-	Underclearance, angles of approach and departure.	4.6.2.7	3.9.15
X	X	-	-	Upright tilt, tilt time tilt cavitation.	4.6.2.8	3.9.10
X	X	X	-	Drift.	4.6.2.9	3.9.12
X	X	-	-	Longitudinal gradeability.	4.6.2.10	3.9.3, 3.10.2.8
X	X	-	-	Dynamic stability, parking brake, emergency brake.	4.6.2.11	3.9.1.2, 3.17.1.3, 3.17.1.4, 3.17.2.2, 3.17.2.3
				Steering.	4.6.2.12	-
X	X	-	-	Obstacle course No. 1.	4.6.2.12.1	3.16
X	-	-	-	Steering control.	4.6.2.12.2	3.16.1
X	-	-	-	Transportability.	4.6.2.13	3.1.2, 3.1.2.1, 3.1.2.2
X	-	-	-	ROPS and FOPS.	4.6.2.14	3.5.1
X	-	-	-	Overload.	4.6.2.15	3.13.2.1
X	X	-	-	Driveline stall.	4.6.2.16	3.11.1.1, 3.11.2, 3.24.1
X	X	-	-	High speed.	4.6.2.17	3.9.16
X	X	-	-	Service brake.	4.6.2.18	3.17, 3.17.1, 3.17.2
X	X	-	-	Maintenance.	4.6.2.19	3.7
X	X	-	-	Static stability.	4.6.2.20	3.9.1.1
				Hydraulic system.	4.6.2.21	-
X	-	-	-	Pump inlet pressure.	4.6.2.21.1	3.12.1.1.4
X	-	-	-	Hydraulic filter and filter assembly.	4.6.2.21.2	3.12.1.3.1, 3.12.1.3.2, 3.12.1.3.3
X	X	X	X	Hydraulic system cleanliness.	4.6.2.21.3	3.12.1.9

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Table III. Test Schedule (Cont'd)

1	2	3	4	5	6	7
X	X	-	X	Hydraulic cylinders.	4.6.2.21.4	3.12.1.7
X	-	-	-	Pump contaminant tolerance.	4.6.2.21.5	3.12.1.2
X	-	-	-	Hydraulic fittings.	4.6.2.21.6	3.12.1.1.1
X	X	X	-	Tophandler operations.	4.6.2.22	3.9.11, 3.9.13, 3.9.14, 3.9.19, 3.13.1
X	X	-	-	Fork operations.	4.6.2.23	3.13.2, 3.13.2.1
X	X	-	-	Electromagnetic interference.	4.6.2.24	3.27
X	-	-	-	Slinging provisions	4.6.2.25	3.30
X	-	-	-	Tiedown provisions.	4.6.2.26	3.30
				Noise level measurement.	4.6.2.27	-
X	X	-	-	Operator's station.	4.6.2.27.1	3.6.1
X	X	-	-	Exterior spectator.	4.6.2.27.2	3.6.1
X	X	-	-	Noise hazard.	4.6.2.27.3	3.6.2
X	-	-	-	Rain.	4.6.2.28	3.10.2.8, 3.11.2.1, 3.11.2.2, 3.4.4
X	-	-	-	Low temperature (minus 25° F).	4.6.2.29	3.4.2, 3.4.3, 3.10, 3.10.2.6, 3.12.1
X	-	-	-	Fording.	4.6.2.30	3.9, 3.9.5, 3.10.2.3, 3.10.2.9
X	X	-	-	Towing.	4.6.2.31	3.14, 3.24.2
X	X	-	-	Controls.	4.6.2.32	3.22.6.1, 3.22.7
X	X	-	-	Rear Axle Oscillation.	4.6.2.33	3.11.3
X	-	-	-	Endurance.	4.6.2.34	3.9.17, 3.10.2.4
-	X	-	-	Reliability.	4.6.2.35	3.8, 3.10, 3.10.2.6, 3.11.1, 3.11.2, 3.12.1
-	X	-	-	Maintainability.	4.6.2.36	3.7
X	X	-	X	Run-in.	4.6.2.37	Section 3
X	-	X	-	Post-test.	4.6.2.38	Section 3

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4.6.2.2 Engine. The diesel engine shall be tested or certified in accordance with paragraphs 5.2 and 5.3 of MIL-STD-1410.

4.6.2.3 Lift height, low engagement height, weights and dimensions. Test the truck in accordance with Test Method No. 1. Nonconformance to 3.9.8 and 3.9.9 shall constitute failure of this test.

4.6.2.4 Lifting speed and lowering speed. Test the truck in accordance with Test Method No. 2. Nonconformance to 3.9.6 or 3.9.7 shall constitute failure of this test.

4.6.2.5 Curb clearance circles. Determine the curb clearance circle in forward and reverse as defined in field Test Procedure of SAE J695 with rated load in a 20 foot container in carry position. Nonconformance to 3.9.18 shall constitute failure of this test.

4.6.2.6 Travel speed. Test the truck in accordance with Test Method No. 3. Speeds not as specified in 3.9.4 shall constitute failure of this test. Inability to shift as specified in 3.11.1.1 shall constitute failure of this test. Exhaust back pressure not in accordance with 3.10.2.8 shall constitute failure of this test.

4.6.2.7 Underclearance, angles of approach, and departure. Test the truck in accordance with Test Method No. 4. Nonconformance to 3.9.15 shall constitute failure of this test.

4.6.2.8 Upright tilt, tilt time, tilt, cavitation. Test the truck in accordance with Test Method No. 5. Nonconformance to 3.9.10, shall constitute failure of this test.

4.6.2.9 Drift. Test the truck in accordance with Test Method No. 6. Nonconformance to 3.9.12 shall constitute failure of this test.

4.6.2.10 Longitudinal gradeability. Test the truck in accordance with Test Method No. 7. Inability of truck to ascend the slope as specified in 3.9.3 shall constitute failure of this test. Exhaust back pressure not in accordance with 3.10.2.8 shall constitute failure of this test.

4.6.2.11 Dynamic stability, parking brake, emergency brake. Test the truck in accordance with Test Method No. 8. Nonconformance to 3.9.1.2, 3.17.1.3, 3.17.1.4, 3.17.2.2, or 3.17.2.3 shall constitute failure of this test.

4.6.2.12 Steering.

4.6.2.12.1 Obstacle course No. 1. Test the truck in accordance with Test Method No. 9. Inability of truck to negotiate the test course or hitting of any markers shall constitute loss of operator control specified in 3.16 and shall constitute failure of this test.

4.6.2.12.2 Steering control. Test the steering control in accordance with Test Method No. 10. Nonconformance to 3.16.1 shall constitute failure of this test.

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4.6.2.13 Highway marine and rail transportability. Test the truck in accordance with Test Method No. 11. Nonconformance to 3.1.2.1, 3.1.2.2, or 3.1.2.3 shall constitute failure of this test.

4.6.2.14 ROPS and FOPS. The ROPS and FOPS shall be tested in accordance with SAE J1040 and SAE J1043. Nonconformance to 3.5.1 shall constitute failure of this test.

4.6.2.15 Overload. When forks are specified test the truck in accordance with Test Method No. 12. Nonconformance to 3.13.2.1 shall constitute failure of this test.

4.6.2.16 Driveline stall. Test the truck in accordance with Test Method No. 13. Nonconformance to 3.11.1.1, 3.11.2 or 3.24.1 shall constitute failure of this test.

4.6.2.17 High Speed. Test the truck in accordance with Test Method No. 14. Nonconformance to 3.9.16 shall constitute failure of this test.

4.6.2.18 Service brake. Test the truck in accordance with Test Method Nos. 15, 16, and 17. Nonconformance to 3.17, 3.17.1, or 3,17.2 shall constitute failure of this test.

4.6.2.19 Maintenance. Test the truck in accordance with Test Method No. 18. Nonconformance to 3.7, as applicable shall constitute failure of this test.

4.6.2.20 Static stability. Test the truck in accordance with Test Method No. 19. Nonconformance to 3.9.1.1 shall constitute failure of this test.

4.6.2.21 Hydraulic systems

4.6.2.21.1 Pump inlet pressure. Test each pump in accordance with Test Method No. 20. Pressure differential between atmospheric pressure and the vacuum in the line in excess of that specified in 3.12.1.1.4 shall constitute failure of this test.

4.6.2.21.2 Hydraulic filter and filter assembly. Test the hydraulic filter in accordance with Test Method No. 21. Nonconformance to 3.12.1.3.1, 3.12.1.3.2, or 3.12.1.3.3 as applicable, shall constitute failure of this test.

4.6.2.21.3 Hydraulic system cleanliness. Test each truck in accordance with Test Method No. 22. Continue cleanup procedures until the average of not less than three consecutive counts conform to 3.12.1.9.

4.6.2.21.4 Hydraulic cylinders. Test each cylinder in accordance with Test Method No. 23. Nonconformance to 3.12.1.7 shall constitute failure of this test.

4.6.2.21.5 Pump contaminant tolerance. Test the pump(s) in accordance with procedure set up NFPA Recommended Standard T.3.9.18-1976. Using the test

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results with the computer program furnished by the procuring activity (see 6.8), calculate and present a 1000 hour Contaminant Tolerance Profile on the particulate contamination chart shown on Figure 1. The profile nonconformance to 3.12.1.2 shall constitute failure of this test.

4.6.2.21.6 Hydraulic fittings. Test the truck in accordance with Test Method No. 24. The contractor shall furnish a complete set of hose assemblies with reusable fittings for conduct of this test. Nonconformance to 3.12.1.1.1 shall constitute failure of this test.

4.6.2.22 Tophandler operations. Test the truck in accordance with Test Method No. 25. Nonconformance to 3.9.11, 3.9.13, 3.9.14, 3.9.19, or 3.13.1 shall constitute failure of this test.

4.6.2.23 Fork operations. When forks are specified (see 6.2), test the truck in accordance with Test Method No. 26. Nonconformance to 3.13.2 or 3.13.2.1 shall constitute failure of this test.

4.6.2.24 Electromagnetic interference. The truck shall be tested in accordance with MIL-STD-461. Nonconformance to 3.27 shall constitute failure of this test.

4.6.2.25 Slinging provisions. Test the slinging provisions by applying the load for the time period specified in MIL-STD-209. Nonconformance to 3.30 shall constitute failure of this test.

4.6.2.26 Tiedown provisions. Test the tiedown provisions by applying the ultimate load for the time period specified in MIL-STD-209. Nonconformance to 3.30 shall constitute failure of this test.

4.6.2.27 Noise level measurement.

4.6.2.27.1 Operator's station. Measure cumulative noise exposure (equivalent continuous noise level, Leq*) at the operator's station in accordance with MIL-STD-1474 or SAE J1166. Truck operations in accordance with Test Method No. 34 shall be used in lieu of the steady-state test of MIL-STD-1474 or in lieu of the bucket loader simulated work cycle of SAE J1166. All lifting and lowering operations shall be accomplished with the truck stationary at maximum lifting and lowering speed. Cumulative noise exposure measurement shall require a minimum of 1 hour of continuous operation on the test course. Nonconformance to 3.6.1 shall constitute failure of this test.

4.6.2.27.2 Exterior spectator. Measure the exterior noise level in accordance with SAE J88. Nonconformance to 3.6.1 shall constitute failure of this test.

4.6.2.27.3 Noise hazard. Measure the noise level at the operator's station and on a 5-foot perimeter while operating the truck at maximum governed speed raising rated load. If the noise level exceeds 85dB(A) at any location, noise hazard warning signs and the appropriate discussion of noise hazard specified in 3.6.2 shall be provided.

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4.6.2.28 Rain. Test the truck in accordance with Test Method No. 27. Nonconformance to 3.4.4, 3.10.2.8, 3.11.2.1, or 3.11.2.2 shall constitute failure of this test.

4.6.2.29 Low temperature (minus 25° F). Test the truck in accordance with Test Method No. 28. One or more of the following shall constitute failure of this test:

(a) Damage to any part of the truck due to storage at minus 30° F (3.4.3).
(b) Engine does not start within 5 minutes after beginning the first cranking (see 3.4.2).
(c) Engine does not run smoothly within 15 minutes after starting without unnatural or continued control manipulation and without use of primers or priming aids (3.4.2).
(d) Instruments and gages do not operate (3.4.2).
(e) Battery charging system does not show positive charge (3.4.2).
(f) With the operator wearing arctic mittens and boots:

(1) Power train cannot be engaged to the wheels (3.4.2).
(2) Any container handling function cannot be operated through two complete movement cycles (3.4.2).

(g) operator encounters difficulty in mounting to and dismounting from the operator's position on the truck (3.4.2).
(h) Measured temperatures of engine oil (3.10), of engine coolant (3.10.2.6), or hydraulic oil (3.12.1) not within specified limits.

4.6.2.30 Fording. The truck shall be tested in accordance with Test Method No. 29. Any of the following shall constitute failure of this test:

(a) Evidence of water damage or malfunction of the torque converter, engine oil sump, transmission, transfer case, fuel tank, hydraulic reservoir, either axle, any wheel, instruments, or gages (3.9.5) or electric drivetrain components (as applicable).
(b) Malfunctioning of any electrical component (3.9.5).
(c) Malfunctioning of any brake (3.9.5).
(d) Malfunctioning of any hydraulic component (3.9.5).
(e) Malfunctioning of the steering system (3.9.5).
(f) Evidence of corrosion which affects the correct functioning of the truck (3.9.5).
(g) Water entering the engine (3.10.2.9).
(h) Inability to start the engine in the water (3.10.2.3).
(i) Volumetric water content in excess of that specified (3.9.5).

4.6.2.31 Towing. Test the truck In accordance with Test Method No. 30. Nonconformance to 3.14 or 3.24.2 shall constitute failure of this test.

4.6.2.32 Controls. Test the controls in accordance with Test Method No. 31. Nonconformance to 3.22.6.1 or 3.22.7 shall constitute failure of this test.

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4.6.2.33 Rear axle oscillation. Test the truck in accordance with Test Method No. 32. Nonconformance to 3.11.3 shall constitute failure of this test.

4.6.2 34 Endurance. Test the truck in accordance with Test Method No. 33 for 1400 cycles in not more than 400 hours. Each tophandler shall be tested for not less than 450 cycles. Nonconformance to 3.9.17 or 3.10.2.4 shall constitute failure of this test.

4.6.2.35 Reliability. Using the MTBF and cycle rate specified in 3.8, the truck(s) shall be tested in accordance with Test Method No. 33 for 400 hours total operating time. More than five relevant failures shall constitute failure of this test. For the purpose of determining MTBF, a failure is defined as any malfunction which causes any unscheduled maintenance action to be taken in order to restore the truck to specified requirements. Those malfunctions resulting from accidents, noncompliance with prescribed operational or maintenance procedures, faulty test instrumentation, fail-safe devices operating as specified, and dependent secondary failures shall be considered nonrelevant failures. Nonrepetitive replacement of light bulbs, fuses, tires, and tightening of fasteners shall be considered nonrelevant failures. Any impending malfunction detected which constitutes a safety hazard to operating personnel or would cause secondary damage either to the item under test or testing instrumentation will be corrected prior to malfunction and considered as a relevant failure. Hours specified herein shall be productive clock hours excluding fueling, servicing, and downtime. In addition, exceeding the temperature limits specified for the cooling system (3.10.2.6), engine oil (3.10), transmission oil (3.11.1), universal joints (3.11.1), hydraulic system 3.12.1, or electric drive train (3.11.2) shall constitute failure of this test.

4.6.2.36 Maintainability. Maintenance times shall be recorded throughout testing in accordance with 4.6.2.35. A maintenance ratio of more than 0.35 as specified in 3.7 shall constitute failure of this test.

4.6.2.37 Run-in. Each truck shall be driven through a run-in for not less than 1 hour. The run-in shall include operating in all gears, operating all electrical equipment, and performing all container handling functions with rated load as well as no load. Evidence of any deficiency, need for adjustment or maintenance, malfunction of any component, or evidence of leakage of air, fuel, lubricant or hydraulic oil, shall be cause for rejection of the truck by the Government until corrected to the Government's satisfaction. The Government may require additional run-in after the deficiency is corrected.

4.6.2.38 Posttest disassembly and examination. Upon completion of all tests, measure and record the compression in each engine cylinder and dismantle the truck sufficiently to permit detailed disassembly and visual examination of the following components:

(a) Engine cylinder heads, all valves, two pistons and piston pins taken from the two cylinders having the lowest compression reading at the end of the test, one main bearing and one connecting rod bearing.

(b) Torque converter and transmission.

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(c) Universal joints and drive shafts.
(d) Differential.
(e) Front axle bearings and seals.
(f) Complete brake system components.
(g) Steering system cylinders, valves, and pump.
(h) Hydraulic system pump, valves, one cylinder of each size, control levers, and any hose and fitting where leakage occurred during the tests.
(i) Steer axle bearings and seals.
(j) Mast assembly bearings and chains.
(k) Electric drive train components.

Additional disassembly and examination shall be made at the option of the contracting officer. The truck shall be examined in accordance with Table II, Column 1. Record complete description of the condition of all parts and assemblies examined, and take pictures of any deficiency. Each evidence of corrosion which affects the functions of the truck or evidence of permanent deformation, breakage, or excessive wear of any component shall constitute a defect. Manufacturing tolerances listed in the contractor's quality control program shall be made available upon request and shall be used as the basis for determining excessive wear.

4.7 Inspection comparison. The Government may select trucks at any time during the contract production period and subject these trucks to the examination specified in 4.6.1 and Column 1 of Table II, and to the tests marked "X" in Column 1 of Table III to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected first article forklifts. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison truck to meet any requirement specified herein, during and as a result of the examination and tests specified in 4.7, shall be cause for rejection of the inspection comparison truck(s) and shall be cause for refusal by the Government to continue acceptance of production trucks until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on trucks previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all trucks accepted prior to the completion of inspection comparison are similarly deficient, unless evidence to the contrary is furnished by the contractor, and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging. The preservation, packing, and marking shall be examined to determine compliance with the requirements of Section 5 and with Appendix II of MIL-STD-162.

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5. PACKAGING

5.1 Preservation. Preservation shall be Level A, B, or C as specified (see 6.2).

5.1.1 Level A. Each complete truck shall be preserved in accordance with the Level A preservation and packaging requirements of MIL-STD-162, with the following exception.

5.1.1.1 Disassembly. The truck shall be sectionalized as recommended by the contractor to meet shipping regulations into sections such as but not limited to, the chassis, personnel cab, ROPS and FOPS if not an integral part of the cab, mast, and spreader. Additional disassembly shall be the minimum necessary to safeguard parts vulnerable to damage, pilferage, and loss, and to the removal of protruding parts that would otherwise significantly increase cubage. Removed bolts, nuts, screws, pins, and washers shall be coated with Type P-1 preservative and then reinstalled in mating parts and secured to prevent loss, or placed in fiberboard boxes conforming to PPP-B-636, W5c or W6c, as applicable to the weight of contents. Boxed items shall be segregated in a manner to allow easy identification.

5.1.2 Level B. Each complete truck shall be preserved in accordance with the Level B preservation and packaging requirements of MIL-STD-162, with the exception specified in 5.1.1.

5.1.3 Level C. Each complete truck, and all items issued with the truck, shall be disassembled in accordance with 5.1.1.1 and then suitably preserved to afford protection against deterioration and damage from the contractor to the initial destination. In addition, the following specific requirements shall be accomplished. Items with critical surfaces required for fit or function shall be provided protection by the use of preservative coating, volatile corrosion inhibitors, or desiccated packs. Items requiring protection from physical damage to surfaces, or which are fragile by nature, shall be protected by wrapping, pack compartmentization, or cartonizing of the individual item. The cooling system shall be filled with a fresh clean solution of 50 percent water and 50 percent antifreeze conforming to MIL-A-46153, and a tag attached indicating the temperature to which the cooling system can be subjected before damage. The crankcase shall contain the required amount of lubricating oil. A guide shall be furnished indicating in detail the preservation materials to be removed, and any other servicing required prior to placing the truck in operation.

5.2 Packing. Packing shall be Level A or C, as specified (see 6.2).

5.2.1 Level A. Each complete disassembly truck, preserved as specified in 5.1, shall be packed in accordance with the Level A packing requirements of MIL-STD-162, Type I or II as recommended by the contractor to meet shipping regulations. Specifically the personnel cab, if removed, shall be packed Type II in accordance with MIL-STD-162.

5.2.2 Level C. Each complete disassembled truck, preserved as specified in 5.1, shall be prepared for shipment as recommended by the contractor but in a

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manner that will assure carrier acceptance and safe delivery to destination at lowest ratings.

5.3 Marking. Packages, shipping containers, and unboxed components shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The truck is intended for use as a container handling truck with the capability of stacking and of loading and unloading from flatbed trailers and rail cars, the 8 foot wide family of commercial and military owned ISO designation 1A and 1C containers and SEALAND containers weighing 50,000 pounds or less. The mobility of the truck, with and without rated load, is intended to cover working areas where high flotation and traction characteristics are required.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When first article preproduction model teats are not required (see 3.2.1).
(c) Time frame required for submission of the first article (preproduction model) and number of trucks required (see 3.2.1).
(d) When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article preproduction model examination and tests, the contracting officer should specify time frame and which examination and tests will be conducted by the Government, and which examination and tests shall be conducted by the contractor (see 3.2.1).
(e) When first article initial production inspection is not required.
(f) Number of initial production trucks to be furnished (see 3.2.2).
(g) Number of trucks to be furnished with tophandlers compatible with 1A containers (40 foot length) (see 3.13.1).
(h) Number of trucks to be furnished with tophandlers compatible with Sea Land containers (35 foot length) (see 3.13.1).
(i) Number of trucks to be furnished with fork kit (see 3.13.2).
(j) Degree of preservation and packing required (see 5.1 and 5.2).

6.3 Preproduction model. Any changes or deviations of production truck from the approved preproduction truck during, production will be subject to the approval of the contracting officer. Approval of the initial production truck will not relieve the contractor of his obligation to furnish trucks conforming to this specification.

6.4 Incident report. The contracting officer should require the contractor to furnish a written report within 24 hours of any incident of equipment malfunction or failure, during the conduct of any preproduction or initial production test. As a minimum, the report should describe components and parts affected, test and operating condictions, date of incident, hourmeter reading, how detected, and description of incident.

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6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

6.6 Noise. When the noise level of the truck exceeds 85 dB(A) the appropriate discussion of the noise hazard per MIL-STD-1474 (MANUALS Paragraph) should include the requirement for hearing protection, the noise level of the truck, the distance from the truck at which 85 dB(A) will always be met, and what operational situations or system configuration will reduce the level to 85 dB(A).

6.7 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the trucks covered by this specification, as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.8 The contracting officer should arrange to furnish the computer program for Contaminant Tolerance Profile required in 4.6.2.21.5.

Custodian:
Army - ME

Preparing activity:
Army - ME

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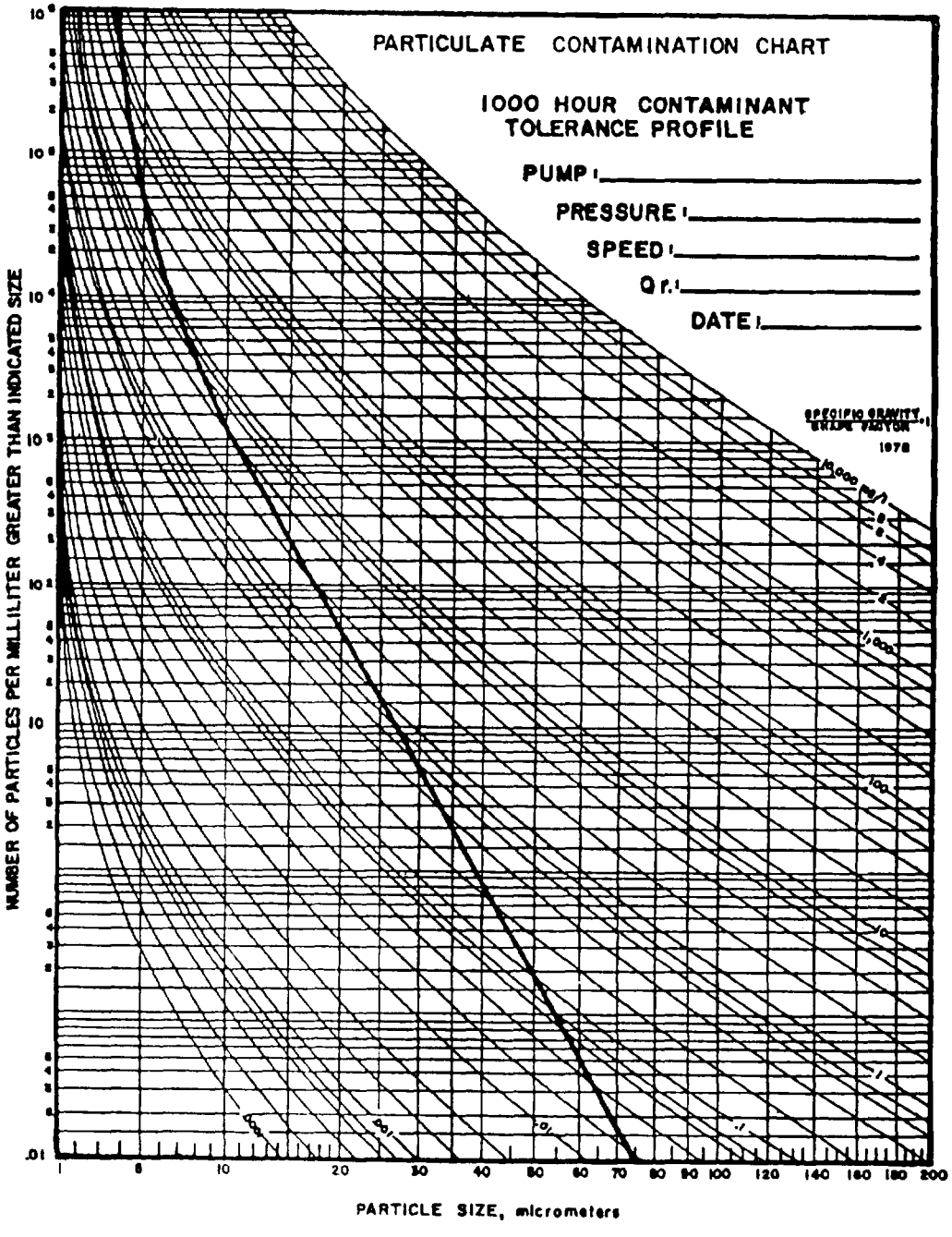


FIGURE I. CHART FOR CONTAMINANT TOLERANCE PROFILE.

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APPENDIX

FOR

TRUCK, CONTAINER HANDLER, WITH TOP LIFT ATTACHMENTS,

DIESEL ENGINE DRIVEN, PNEUMATIC TIRED, FOUR WHEEL

DRIVEN, ARTICULATED FRAME STEER, ROUGH TERRAIN;

50,000 POUND CAPACITY AT 48 INCH LOAD CENTER

SCOPE. The test methods contained in this appendix are to determine whether container handlers procured under this specification conform to the requirements set forth.

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APPENDIX

TEST METHOD NO. 1

LIFT HEIGHT - LOW ENGAGEMENT HEIGHT

WEIGHT AND DIMENSIONS

1. TEST COURSE:

   (a) Level, flat surface.
   (b) Level, flat surface with sharp drop-off over an edge.

2. TEST APPARATUS:

   (a) Linear measurement equipment.
   (b) Liquid level or clinometer.

3. TEST PROCEDURE:

   (a) Lift the rated load to maximum height obtainable. Adjust the mast tilt to level the load. Measure the true vertical distance from the surface on which the truck is positioned to the bottom surface of the container. (This is the maximum lift height).
   (b) Measure the true vertical distance from the ground surface to the uppermost projection of the upright assembly with mast in true vertical position without load as determined by a clinometer. (This is the collapsed mast height.)
   (c) Place rated load on a level surface which is a minimum of 12 inches below the level ground surface on which the truck is positioned. Measure the true vertical distance between the surface on which the container is positioned and the surface on which the truck is positioned (this is the low engagement height). Engage load and lift to a minimum of 3 feet above ground level.
   (d) Repeat (a) and (c) with rated load in a 20 foot container.
   (e) With mast in lowest position, measure the height of the unloaded truck in accordance with SAE J894.
   (f) With top handler removed, measure the width of the unloaded truck in accordance with SAE J894.
   (g) With the top handler removed, measure the length of the unloaded truck in accordance with SAE J894.
   (h) Weigh the truck fully fueled and serviced, without load. Obtain front and rear axle weights with and without rated load in carry position (see 3.1.1.1), for use in calculating the Flotation Index, The truck shall be in the straight ahead position.

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APPENDIX

TEST METHOD NO. 2

LIFTING SPEED - LOWERING SPEED

1. TEST APPARATUS:

(a) Linear measurement equipment.
(b) Stopwatch.

2. TEST PROCEDURE:

(a) Record time in seconds required to raise rated load from ground to maximum lift height (with mast vertical) at maximum engine speed.
(b) Record time to lower rated load at maximum speed (full open valve) to 3 to 4 foot height. Load to be abruptly stopped at the 3 to 4 foot height. Record height at stop. Engine to be at low idle speed for hydraulic mechanical drive truck.
(c) Repeat steps (a) and (b) a minimum of 3 times.
(d) Compute lifting speed and lowering speed and use the fastest rate obtained from each as the established speeds.
(e) Remove load and raise empty top handler to maximum height. Record time in seconds required to lower top handler to lowered position at maximum speed (full open valve) and engine at low idle speed for hydraulic mechanical drive truck. Repeat 3 times.
(f) Compute the lowering speed and use the fastest rate as the established speed.
(g) When forks are furnished, remove top handler and perform 2(e) and 2(f) with the forks.

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APPENDIX

TEST METHOD NO. 3

TRAVEL SPEED

1. TEST COURSE:

   (a) The test course shall be a dry, level surface of sufficient length to attain maximum speed prior to entering measured distance, plus measured distance, plus sufficient stopping distance.

2. TEST APPARATUS:

   (a) Linear measurement equipment.
   (b) Stopwatch or electronic timer.
   (c) Pressure measurement device.

3. TEST PROCEDURE:

   (a) Install exhaust system pressure measurement device not more than 2 inches beyond the header outlet.
   (b) Operate the truck at governed engine speed rearward with rated load in a 20 foot container.
   (C) Drive the truck to attain maximum speed prior to entering the measured test course (do not exceed an operating speed at which the truck cannot be controlled). While accelerating, perform gear or range changes under full throttle. Loads shall be in the carry position.
   (d) Drive through the measured distance at maximum speed, recording the length of the distance and the time required to traverse it and the exhaust pressure. The measured distance shall be not less than 88 feet.
   (e) Turn the truck around and repeat the test approaching the measured distance from the opposite end. Repeat for a total of six runs, three in each direction through the measured distance. Travel speed shall be the average of the six runs.
   (f) Repeat (c) thru (e) except truck shall be driven forward without load at governed engine speed.

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APPENDIX

TEST METHOD NO. 4

UNDERCLEARANCE, ANGLES OF APPROACH AND DEPARTURE

1. TEST COURSE:

   (a) Level, flat surface.

2. TEST APPARATUS:

   (a) Linear measurement equipment.
   (b) Triangular template of 20°.

3. TEST PROCEDURE:

   (a) Tires shall be inflated to the pressure specified by the tire manufacturer for maximum tire loading at 2-1/2 mph rating.
   (b) Measure the vertical distance from the surface to the lowest point on the vehicle, excluding wheels and mast.
   (c) Raise top handler 12 inches minimum and tilt mast to full back tilt. Measure the clearance under the mast.
   (d) Using the triangular template, determine if the angles of approach and departure exceed those specified.

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APPENDIX

TEST METHOD NO. 5

UPRIGHT TILT - TILT TIME -

TILT CAVITATION

1. TEST COURSE:

   (a) Level, flat surface.

2. TEST APPARATUS:

   (a) Clinometer.
   (b) Stopwatch or electronic timer.
   (c) Hydraulic pressure gage capable of reading below atmospheric pressure.
   (d) Thermocouple.

3. TEST PROCEDURE:

   (a) Upright tilt.

      (1) Place truck without load on level, flat surface, using clinometer to insure truck is level.
      (2) Raise top handler to a height of 10 feet above the surface and tilt mast to most forward container handling operating position, and record angle on clinometer located on front surface of outer mast channel.
      (3) If tilt override is furnished, repeat 3(a)(2) after override is energized.
      (4) Tilt mast as far rearward as possible and record angle of tilt on clinometer located on front surface of outer mast channel.

   (b) Tilt time.

      (1) Anchor rear of truck with external means. Lift the rated load to maximum lift height.
      (2) Tilt the load to maximum forward tilt without adjustment of height control.
      (3) At governed engine speed tilt the load from maximum forward tilt, to maximum rearward tilt, and record the time required.
      (4) Repeat the procedure tilting from rearward to forward at governed engine speed. And record the time required.

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APPENDIX

TEST METHOD NO. 5 (CONT'D)

(c) Tilt cylinder cavitation.

(1) Install the pressure gags to read hydraulic pressure at the head end of the tilt cylinder. Install thermocouple to read oil temperature at the hydraulic tank.
(2) After heating the oil to specified temperature, lift the rated load to maximum lift height.
(3) With the engine at idle speed, slowly tilt the load from full rearward to full forward and back again. Tilt forward shall be done with control valve wide open 5 times and "feathered" 5 times. Observe and record any evidence of cavitation in the tilt circuit or loss of operator control.
(4) Record lowest pressure reading at tilt cylinder and oil temperature reading.

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APPENDIX

TEST METHOD NO. 6

DRIFT

1. TEST COURSE:

   (a) Level, flat surface.

2. TEST APPARATUS:

   (a) Temperature thermocouple and potentiometer or equivalent temperature measuring device.
   (b) Clinometer.

3. TEST PROCEDURE:

   (a) Install the thermocouple in the volumetric center of the hydraulic-system oil reservoir. Heat the hydraulic oil by exercising hydraulic functions until the oil temperature exceeds 120° F. Maintain oil temperature above 120° F for the duration of the test.
   (b) Raise rated load to maximum lift height. Tilt mast to true vertical position measured with the clinometer. Mark reference point on mast.
   (c) Hold the load for 2 minutes and record the downward drift rotational drift, and hydraulic temperature in that period.

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APPENDIX

TEST METHOD NO. 7

LONGITUDINAL GRADEABILITY

1. TEST COURSE:

   (a) The length of the test course shall be not less than three times the wheelbase of the truck, graded to 25 percent within plus or minus 1 percent grade. Grade resistance shall be sufficient to prevent observable wheel slip. Test course surface shall be unpaved.

2. TEST APPARATUS:

   (a) Speed indicator.
   (b) Exhaust system pressure measurement device.

3. TEST PROCEDURE:

   (a) Install exhaust system pressure measurement device not more than 2 inches beyond the header outlet. Record exhaust pressure for each run.
   (b) Drive the truck with rated load in carry position forward up the slope, entering the course at a speed of not leas than 2 mph, nor more than 4 mph. Continue through the course, maintaining a speed of not leas than 2 mph. Make a total of three such passes up the slope.

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APPENDIX

TEST METHOD NO. 8

DYNAMIC STABILITY -

PARKING BRAKE - EMERGENCY BRAKE

1. TEST COURSE:

(a) The length and width of the test course shall be not less than the truck curb clearance circle diameter, graded to 15 percent slope, within plus or minus 1 percent grade. Test course surface shall be unpaved.
(b) The length of the test course shall be sufficient to conduct the emergency brake tests, graded to 25 percent within plus or minus 1 percent grade. Test course surface shall be unpaved.
(c) Test course shall be a dry, level, improved surface allowing no penetration, with a grade not exceeding plus or minus 1 percent.

2. TEST APPARATUS:

(a) Force measurement equipment.
(b) Linear measurement equipment.
(c) Speed indicator.

3. TEST PROCEDURE:

(a) Full circle performance.

(1) Drive the truck with rated load side shifted to extreme left position (oscillation centered) onto the slope specified in 1(a). Perform a complete forward circle at maximum left turn (without steering override) while on the slope.
(2) Repeat 3(a)(1) with load side shifted to extreme right position.
(3) Repeat 3(a)(1) except perform a right full turn.
(4) Repeat 3(a)(1) except perform a right full turn with load side shifted to the extreme right position.
(5) Repeat 3(a)(1) through 3(a)(4) without rated load.

(b) Slope descension, braking, and parking brake operation.

(1) Position the truck with rated load in carry position at the top of the slope specified in 1(b).
(2) Drive the truck forward down the slope until a speed of not less than 2 mph is attained, then stop and hold the truck for not less than 1 minute.

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APPENDIX

TEST METHOD NO. 8 (CONT'D)

(3) Lower the load to ground level. Back away from the load until the truck clears the load, and stop the truck.
(4) Apply the parking brake. Measure and record the force necessary to apply the parking brake at the center of the handgrip.
(5) Turn the truck around so that it is facing forward up hill. Repeat 3(b)(4).

(c) Emergency brake operation.

(1) Air or hydraulic powered brakes.
Position the truck with rated load in carry position at the top of and facing rearward down the slope specified in 1(b). Shut off the engine and allow the truck to roll down the slope in neutral until a speed of 2 mph is attained. Apply the emergency brake provisions (parking or service brake as applicable) and bring the truck to a stop. Release the brake, allow the truck to begin to roll, and apply the brake again to stop the truck and repeat for a total of six dead engine stops.

(2) Electric brakes.
Drive the truck with rated load in carry position over the test course used for service brake tests (Test Method No. 15). When the truck reaches a minimum 9 mph, apply the emergency brake system and the regenerative system together. Measure and record the distance between the point of application and the point of stoppage.

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APPENDIX

TEST METHOD NO. 9

STEERING

1. **TEST COURSE:**

   (a) The test course shall be a dry, level surface in accordance with Figure A-1.

2. **TEST APPARATUS:**

   (a) Speed indicator.

3. **TEST PROCEDURE:**

   (c) Obstacle course in accordance with Figure A-1.

   (1) Drive the truck forward, without load over the test course twice, once in each direction, at the truck's maximum capable travel speed.
   (2) Drive the truck in reverse, with rated load, over the test course, once in each direction, at a speed of not less than 9 mph. Load shall be raised to a height to clear course markers.

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APPENDIX

TEST METHOD NO. 10

STEERING CONTROL

1. **TEST COURSE:**

   (a) Smooth, dry, level concrete surface.

2. **TEST APPARATUS:**

   (a) Force measurement equipment.
   (b) Stopwatch.
   (c) Linear measurement equipment.

3. **TEST PROCEDURE:**

   (a) Place truck without load on dry, level, smooth concrete with wheels in a straight ahead position.
   (b) With the engine operating at low idle, determine the time, and the tangential force at the steering wheel rim, required to turn the wheel from stop to stop in both directions. Repeat 2 more times.
   (c) Repeat 2(b) except that engine shall operate at maximum governed speed.
   (d) Count the number of revolutions required to turn the wheel from stop to stop in each direction. Repeat 2 more times.
   (e) Apply a minimum 300 pound force downward perpendicular to the rim of the steering wheel approximately midway between two spokes and hold for a minimum of 10 seconds.
   (f) Repeat 2(d) except the force shall be applied upward.
   (g) With engine at low idle, apply the maximum force attainable up to 150 pounds, to the rim of the steering wheel, and hold for a minimum of 10 seconds.
   (h) Repeat 2(f) in the opposite direction.
   (i) Measure and record seat to steering wheel clearance, directional control to steering wheel clearance, and steering wheel diameter.

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APPENDIX

TEST METHOD NO. 11

TRANSPORTABILITY.

1. TEST APPARATUS:

(a) Semi-trailer(a) conforming to MIL-S-62164 (HET Semi-trailer) and suitable towing tractor(s).
(b) Cranes of sufficient capacity to handle disassembled sections of vehicle.
(c) LCU-1466 lighter.

2. TEST PROCEDURE:

(a) Highway Transport.

(1) Using the procedures contained in the transportability report furnished by the contractor, disassemble the truck to highway transport configuration.
(2) Record time for disassembly, equipment and manpower utlized, and weight of each major component disassembled.
(3) Load each transport load aboard the semi-trailer.
(4) Record height, width, length, and total weight of each transport load.
(5) Record total time for disassembly and loading, not including time allotted for weighing.
(6) When reassembling truck, record total time for reassembly, equipment and manpower utilized.

(b) Marine Transport.

(1) Drive truck without tophandler on the LCU-1466 lighter.
(2) Drive truck off LCU-1466 lighter.
(3) Note any clearance problems encountered when driving on or off the LCU-1466 lighter.

(c) Rail Transport (rail hump).

(1) The truck with mast and tophandler shall be secured to the rail car using incorporated tiedown provisions.
(2) Hump the rail car, with the truck tied down at a speed of 8 mph against a minimum two buffer cars. Total buffer load excluding car weights shall be 140,000 pounds minimum.
(3) Repeat hump at 8 mph with truck in other direction. The truck may be reversed by either reloading it in the specified direction on the flat car or by reversing the direction of the flat car.
(4) After each impact, the truck and tiedowns shall be inspected to determine the amount of load shift, permanent deformation to components, or other evidence of possible failure.

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APPENDIX

TEST METHOD NO. 12

FORK OVERLOAD

1. TEST COURSE:

(a) Level flat surface.
(b) Anchor.

2. TEST APPARATUS:

(a) Magnetic particle or dye penetrant capability.

3. TEST PROCEDURE:

(a) Place truck on level surface and support it with blocks under the axle or frame to relieve strain on tires. Position mast in true vertical position, with rated load on forks in normal carrying position. External means may be utilized to hydraulically lock the tilt cylinder.
(b) Secure truck to floor in a manner to compensate for the additional test loads to be placed on forks. Counterweight may be added to assist the holding force. Apply counterbalancing forces in a vertical direction.
(c) With crane or another truck, place rated load in position over the first load such that the center of mass of the test load is on the truck longitudinal centerline and at the rated load center.
(d) Check strain or holddown devices and hydraulic system to determine whether the additional load can be placed on forks. Tighten holddown devices if necessary.
(e) With crane or another truck, place the third rated load carefully in position over the first two loads such that the center of mass of load is on the longitudinal centerline of truck and at the rated load center.
(f) When placing the overload on trucks, personnel should stand clear and crane operator should stand by to remove load if there is any evidence of immediate collapse or failure in truck structure.
(g) If truck accepts the additional test load without immediate failure, let overload stand on truck for 10 minutes.
(h) Remove loads.
(i) Inspect truck structure, frame, mast assembly, the hydraulic system for evidence of deformation, fractures, leakage in hydraulic system, broken welds, etc.
(j) Inspect forks a minimum of 6 inches each side of heel by magnetic particle or dye penetrant method.
(k) In lieu of (a) through (h), forks may be tested by application of a force equal to 1.5 times rated load to each fork with a hydraulic ram at 48 inches from heel of fork.

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APPENDIX

TEST METHOD NO. 13

DRIVE LINE STALL

1. TEST APPARATUS:

(a) Timing device.
(b) Thermocouple and potentiometer or equivalent temperature measuring device.
(c) Tachometer.
(d) Thermometer.

2. TEST PROCEDURE:

(a) Mechanical Drive Line.

(1) Install thermocouple to measure oil temperature at the outlet from the converter.
(2) Thermostat shall not be blocked open.
(3) With the ambient temperature not less than 25° F, operate the truck until the torque converter oil outlet temperature is a minimum of 157° F.
(4) Attach the truck pintle hook to a fixed object to prevent forward motion of the truck by a horizontal force on the pintle hook. Shift into the highest forward gear and stall the torque converter for not less than 30 seconds while the engine is operating at governed speed. The transmission disconnect control shall not be used during this test.
(5) Record initial torque converter oil temperature and maximum oil temperature after 30 seconds stall, governed speed at stall, and ambient temperature.

(b) Electric Drive Line.

(1) Install thermocouples to measure temperature on generator, switch gear contacts and all drive motors.
(2) Attach the truck pintle hook to a fixed object to prevent forward motion of the truck by a horizontal force on the pintle hook. Shift to forward speed and stall the truck for not less than 30 seconds.
(3) Record initial temperature measured and maximum temperature after 30 seconds stall.

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APPENDIX

TEST METHOD NO. 14

HIGH SPEED

1. TEST COURSE:

   (a) The test course shall be compacted surface of sufficient size to allow continuous travel at wide open throttle in highest gear range.

2. TEST APPARATUS:

   (a) Speed Indicator.
   (b) Equivalent temperature measuring device.

3. TEST PROCEDURE:

   (a) Block thermostat in the open position.
   (b) Install thermocouples in the hydraulic system reservoir near the pump inlet, in the transmission sump, in the transfer case sump, in the engine coolant outlet, in the torque converter outlet line adjacent to the torque converter, engine oil sump, generator and drive motor (for electric drive).
   (c) Record all temperature including ambient before starting test.
   (d) Drive the unloaded truck over the test course for 2 hours at wide open throttle in highest forward gear range. Stop the truck and record all temperatures immediately, including ambient and universal joint temperatures. Check for universal joint temperature using temperature marker.
   (e) In not leas than 5 minutes after stopping, drive the truck for an additional 2 hours as in (d), and record all temperatures immediately upon stopping, including ambient and universal joint temperatures.

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APPENDIX

TEST METHOD NO. 15

SERVICE BRAKE

1. TEST COURSE:

(a) The test course shall be a dry, level, improved surface allowing no penetration, with a grade not exceeding plus or minus 1 percent.

2. TEST APPARATUS:

(a) Linear measurement equipment.
(b) Detonator, brake apparatus, with powder loads, or equivalent apparatus.
(c) Speed indicator.
(d) Pedal force gage.

3. TEST PROCEDURE:

(a) Stopping distance: Hydraulic mechanical drive.

(1) For air or air over hydraulic brakes, discharge the reservoir completely. Record time required to recharge the system.
(2) Attach the pedal force gage to the brake pedal in a manner which enables force to be applied to the face of the gage.
(3) Connect the detonator to be operated by a switch directly connected to the brake pedal so that it to actuated during the first 1/4 inch of pedal travel. The service brake, without transmission disconnect, shall be used for this test.
(4) Drive the truck with rated load in carry position over the test course in forward direction. When the truck reaches minimum 9 mph, apply the service brakes with no more than 90 pounds pedal force.
(5) Measure and record the pedal force and the distance traveled between the point of brake pedal application and the point of stoppage.
(6) Drive the truck with rated load in carry position over the test course in rearward direction. When the truck reaches minimum 9 mph, apply the service brakes with no more than 90 pounds pedal force.
(7) Measure and record the pedal force and the distance traveled between the point of brake pedal application and the point of stoppage.
(8) Perform 3(a)(4) through (3(a)(7) four times in forward gear and four times in reverse gear; two times in each direction on the course. Observe whether the front wheels lock or any wheel loses contact with the test course. Measure the amount of fishtail.

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APPENDIX

TEST METHOD NO. 15 (CONT'D)

(b) Stopping distance: Electric drive.

(1) Drive the truck with rated load in carry position over the test course in forward direction. When the truck reaches a minimum of 9 mph, stop the truck with the regenerative braking system.
(2) Measure and record the distance traveled between the point of brake application and the point of stoppage.
(3) Repeat 3(b)(1) through 3(b)(2) in the reverse direction.
(4) Repeat 3(b)(1) through (3(b)(3) four times in the forward gear and four times in reverse gear; two times in each direction on the course. Observe whether the front wheels lock or any wheel loses contact with the test course. Measure the amount of fishtail.

(c) Panic stopping: (Hydraulic mechanical and electric drive).

(1) Drive the truck with rated load in carry position in reverse direction over the test course.
(2) When the truck obtains maximum speed, perform a panic stop.
(3) Observe whether any wheel loses contact with the test course. Measure the amount of fishtail.
(4) Perform 3(c)(1) through 3(c)(3) three times.
(5) Disengage load. Perform 3(c)(2) through 3(c)(3) in forward direction 3 times.
(6) Perform 3(c)(2) through 3(c)(3) in reverse direction 3 times.
(7) For hydraulic mechanical drive, measure and record pedal force for all panic stops.

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APPENDIX

TEST METHOD NO. 16

BRAKE FADE

1. TEST COURSE:

   (a) The test course shall be a dry, improved surface allowing no penetration, with grades not exceeding plus or minus 1 percent.

2. TEST APPARATUS:

   (a) Speed indicator.
   (b) Decelerometer.

3. TEST PROCEDURE:

   (a) Drive the truck with rated load up to a minimum of 9 mph and apply the service brakes to produce a deceleration rate of 8 fps per second, plus or minus 1 fps per second, until the truck has slowed to a minimum of 5 mph. Immediately accelerate the truck at full throttle in high gear until the travel speed reaches a minimum of 9 mph. Immediately repeat the test for a total of 35 cycles. On completion of the 35th cycle, accelerate the truck at full throttle in high gear to a minimum 9 mph and immediately conduct a service brake stopping test using procedure in Test Method No. 15. After the brakes have completely cooled, perform the test consisting of 35 snubs and one measured stop a second time. Make no adjustments or repairs between the two test runs.

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APPENDIX

TEST METHOD NO. 17

WET BRAKE

1. TEST COURSE:

   (a) Body of fresh water of sufficient depth to permit submersion of the truck to a minimum 60 inch depth.

2. TEST PROCEDURES:

   (a) Drive the truck into the water to the minimum 60 inch depth. Release the service brakes and allow the truck to stand for not less than 15 minutes. After the 15 minutes have elapsed, drive the truck from the water and immediately conduct a service brake stopping test, using the procedure specified in Test Method No. 15. Service brakes shall not be applied after release in the water until the test application.

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APPENDIX

TEST METHOD NO. 18

MAINTENANCE

1. TEST FACILITY:

(a) The test facility shall consist of a shop area of sufficient size to permit unobstructed surveillance of maintenance.

2. TEST APPARATUS:

(a) Common handtools.
(b) Special tools (if furnished by the contractor with the truck).

3. TEST PROCEDURE:

(a) Examine the truck for accessibility of major assemblies.
(b) Examine position of drains with respect to accessibility.
(c) Using the tools, demonstrate that the major assemblies are accessible for repair and maintenance.
(d) Demonstrate the accessibility of' drains and path of discharge of lubricants and coolants by activating drainage controls.
(e) Perform and time each operation on the maintenance operations list (see 3.7).

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APPENDIX

TEST METHOD NO. 19

STATIC STABILITY

1. TEST COURSE:

   (a) Flat, level surface.

2. TEST APPARATUS:

   (a) 40 foot container loaded to gross weight of 80,000 pounds.
   (b) 40 foot container loaded to rated load.
   (c) Clinometer.

3. TEST PROCEDURE:

   (a) Static Stability - 1.6 times rated load.

      (1) Position truck on test course with steering straight ahead.
      (2) With mast vertical, lift 80,0000 pound test load to minimum 4 foot height above surface on which the truck is positioned. Note any clearance between rear wheels and the surface.

   (b) Static Stability - rated load, full forward tilt.

      (1) Position truck on test course with steering straight ahead.
      (2) Engage rated load and raise to maximum lift height and maximum rear tilt.
      (3) Tilt rated load, while maintaining maximum lift height from full rear tilt to full forward tilt with engine at low idle speed for trucks with hydraulic mechanical drive and at operating speed for trucks with electric drive. Note any clearance between rear wheels and the surface. No feathering of valve shall be allowed during conduct of this test.

      Note: For safety purposes, 3(b) may be conducted above another 40 foot container. Contact of the truck and rated load with the other container is not permitted.

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APPENDIX

TEST METHOD NO. 20

PUMP INLET PRESSURE

1. TEST APPARATUS:

   (a) Pressure gage.
   (b) Hydraulic fittings (as required).
   (c) Thermocouple.

2. TEST PROCEDURE:

   (a) Install a pressure (vacuum) gage in the pump suction line immediately adjacent to the pump (or at the pump) and a thermocouple in the hydraulic tank.
   (b) Heat the oil to specified temperature.
   (c) Run the pump at the maximum speed it can encounter during vehicle operation and record the pump-inlet pressure, and temperature.
   (d) Repeat for each pump installation on the truck.

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APPENDIX

TEST METHOD NO. 21

HYDRAULIC FILTER AND FILTER ASSEMBLY AND CONTAMINATION INDICATOR

1. TEST APPARATUS:

   (a) Hydraulic test stand may be used in place of on vehicle test.
   (b) Pressure gages.
   (c) Temperature gages.
   (d) Flowmeter (when test stand is used).
   (e) Provisions for controlling oil temperature, if necessary.

2. TEST PROCEDURE:

   (a) Pressurize the housing to not less than the minimum proof pressure (100 psi for return line filters, 30 psi for suction line filters, or 1-1/2 times maximum system pressure for pressure filters). Hold the pressure for not leas than 60 seconds and examine for leaks.
   (b) Install a new element in the filter housing. Establish maximum flow through the filter at an oil temperature of 150° F, plus or minus 5° F. Measure and record the pressure drop across the filter assembly.
   (c) Plug the filter element chamber oil exit, or otherwise obstruct all flow through the filter element, leaving the outlet port open. Establish and maintain the oil temperature at 150° F, plus or minus 5° F. Increase the upstream pressure to 10 psi more than determined in (b) or to 2 psi for suction line filters. Observe the contamination indicator. Increase upstream pressure until the indicator enters the bypass area or until bypass flow exceeds 5 percent of the maximum system flow, whichever occurs first. Observe and record this pressure (this is the bypass opening pressure). Increase pressure until the maximum system flow is attained through the bypass valve. Observe and record this pressure.

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APPENDIX

TEST METHOD NO. 22

HYDRAULIC SYSTEM CLEANLINESS

1. TEST APPARATUS:

   (a) Liquid automatic particle counter.

2. TEST PROCEDURE:

   (a) Connect the hydraulic lines at the cylinders to bypass those cylinders whose capacity is less than 1/2 of the connecting line volume.
   (b) Run the engine at no load governed speed pumping the oil through each circuit for not less than 5 minutes. Perform the particle count as described in (d) below.
   (c) When applicable, connect each line to its related component. Operate each function of the system for 5 minutes continuously through full stroke and at maximum rates. Perform the particle count as described in (d) below.
   (d) The procedure for determining contamination shall be as follows.

      (1) Contamination levels shall be determined by particle counts. The particle counter used for evaluation of the samples shall be calibrated in accordance with ANSI Recommended Standard Method for Calibration of Liquid Automatic Particle Counters using "AC" Fine Test Dust (ANSI B93.28). The required counts shall be the average of not less than three consecutive counts. Samples may be taken from the reservoir or upstream of the filter(s) used for cleanup.
      (2) Use any combination of on-the-truck and off-the-truck filtration for cleanup.
      (3) Connect the counter sensor to the system, using a bypass line and a branch line. The bypass line shall connect to the system, have a flow rate not less than 5 times the sensor flow rate, and may return fluid directly to the reservoir. The branch line to the sensor shall tee into the bypass line and shall be not more than 12 inches in length. Any flow restriction devices shall be placed downstream of the particle counter sensor. Start engine and operate at not less than 1000 rpm. Record particle counts.

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APPENDIX

TEST METHOD NO. 23

CYLINDER PROOF

1. TEST APPARATUS:

   (a) Hydraulic test bench or the truck capable of developing two times system working pressure.
   (b) Pressure gage.
   (c) Thermocouple.

2. TEST PROCEDURE:

   (a) Double acting cylinder.

      (1) Position and mechanically hold the piston at the approximate midpoint of the cylinder.
      (2) Fill both sides of the piston with oil at a minimum of 120° F.
      (3) With the head end port capped, apply an oil pressure equal to 1.5 times the working pressure to the rod end of the piston for minimum of 60 seconds. Release pressure.
      (4) With the rod end port capped, apply an oil pressure equal to 1.5 times the working pressure to the head end of the piston for minimum of 60 seconds. Release pressure.

   (b) Single acting cylinder.

      (1) Extend cylinder to maximum extension by filling with oil at a minimum of 120° F.
      (2) Apply oil pressure equal to 1.5 times the working pressure at the inlet port for a minimum of 60 seconds. Release pressure.

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APPENDIX

TEST METHOD NO. 24

HYDRAULIC FITTINGS

1. TEST APPARATUS:

   (a) Common handtools.

2. TEST PROCEDURE:

   (a) Remove one hydraulic hose assembly that has permanent-type fittings.
   (b) Replace the hose assembly with an equivalent field attachable fitting only to assure interchangeability.
   (c) Repeat (a) and (b) above until each hose assembly with permanent-type fittings is replaced with a hose assembly with reusable fittings.
   (d) After every hose assembly with permanent-type fittings has been replaced with a hose assembly with reusable fittings, reinstall the original hose assemblies on the truck.

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APPENDIX

TEST METHOD NO. 25

TOPHANDLER OPERATIONS

1. TEST COURSE:

   (a) Level surface.
   (b) 15 percent slope, within plus or minus 1 percent grade.

2. TEST APPARATUS:

   (a) Clinometer.
   (b) Linear measurement equipment.
   (c) Stopwatch.
   (d) Minimum 8 foot high platform.
   (e) 18 inch high blocks x 8 feet long to support the end of a container.

3. TEST PROCEDURE:

   (a) Oscillation.

      (1) With the truck on a level surface, deposit a 20 foot container with rated load such that one and of container is on top of an 18 inch high block.
      (2) Disengage container, back away, and drive around to other side of container.
      (3) Reengage container and lift to a minimum 5 foot height.
      (4) Repeat (1) through (3) with 35 foot and 40 foot containers.

   (b) Sideshift.

      (1) Position truck with rated load lateral on a 15 percent side slope such that the right side of the truck is uphill.
      (2) Raise rated load to sufficient height to clear the ground.
      (3) Sideshift load to extreme left position.
      (4) Sideehift load to extreme right position and record time and total sideshift from extreme left to extreme right.
      (5) Sideshift load to extreme left position and record time and total sideshift from extreme right to extreme left.
      (6) Turn the truck around and repeat (1) through (5).

   (c) SLEW.

      (1) From one corner of the rated load, draw a line on the pavement such that the line forms a minimum of 3.5.degree angle with the length of the container.
      (2) Align the truck such that the front face of the car riage is above the line.
      (3) Engage the container without backing the truck.
      (4) Repeat 3(d)(1) through 3(d)(3) from opposite corner.

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APPENDIX

TEST METHOD NO. 25 (CONT'D)

(d) Reach.

(1) Lift rated load to minimum 8 foot height.
(2) Deposit load on 8 foot high platform as far waay from leading edge of platform as possible.
(3) Disengage rated load and measure distance from leading edge of platform to the container.
(4) Re-engage rated load and remove from platform.
(5) Adjust mast tilt such that rated load is horizontal. Measure and record the closest horizontal distance between the container and the truck's carriage.
(6) Adjust the load to carry position. Measure and record the closest distance (perpendicular to the container) between the container and the truck's carriage.

(e) Tophandler Interchange.

(1) Remove the 40 foot tophandler from truck and install 20 foot tophandler. Record time and tools required for operator to make the change.
(2) Remove 20 foot top handler from truck and install 35 foot tophandler. Record time and tools required for operator to make the change.

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APPENDIX

TEST METHOD NO. 26

FORK OPERATIONS

1. TEST COURSE:

   (a) Level, flat surface.
   (b) Level, flat surface with sharp drop-off of 12 inches minimum over an edge.

2. TEST APPARATUS:

   (a) Linear measurement equipment.
   (b) Clinometer or liquid level.
   (c) Rigid pole of a length not less than the width of the forklift.
   (d) Stopwatch.

3. TEST PROCEDURE:

   (a) Remove the tophandler and if the fork kit is not already on the truck install the kit without forks. Record time, manpower and procedures to install the kit (if performed). Record time to install the forks.
   (b) Measure the fork length, fork thickness at the heel and at the tip, and the fork taper.
   (c) Place rated load on a level surface which is a minimum of 6 inches below the level ground surface on which the forklift is positioned. Position the forklift on the edge of the drop-off. With mast vertical, lower the forks (without load) over the edge to the full extent of mast travel. Measure the true vertical distance between the surface on which the forklift is positioned and the top surface of the forks (this is low engagement height). Engage rated load with forks and lift to the full extent of mast travel. Adjust the mast tilt to level the load. Measure the true vertical distance from the surface on which the truck to positioned to the top surface of the forks.

   (d) Fork Spread.

      (1) With no load on the forks, move the forks as close together as possible and measure the distance between the outside edges of the forks. Move the forks as far apart as possible and measure the distance over the outside edges of the forks.

   (e) Upright Tilt.

      (1) With the forklift on a level, flat surface without load, raise forks above the surface and tilt as far forward as possible and record angle on clinometer located on front surfaces of outer mast channel.
      (2) Tilt forks as far rearward as possible and record angle of tilt on clinometer.

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APPENDIX

TEST METHOD NO.26 (CONT'D)

(f) Sideshift.

(1) Spread the forks to minimum 90 inches (outside to outside) and center the forks on the truck. Engage 50,000 pound load.
(2) Sideshift forks to extreme right position. Measure sideshift travel.
(3) Sideshift forks to extreme left position and measure travel.

(g) Tire clearance.

(1) Position the rigid pole across the forks and anchor it as far rearward as possible on the horizontal fork surface.
(2) Operate the forks throughout the full range of lift heights and tilt angles observing any interference between the pole and any part of the forklift.

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APPENDIX

TEST METHOD NO. 27

RAIN

1. TEST APPARATUS:

   (a) Timing device.
   (b) Measuring device (U.S. Weather Bureau type preferred).
   (c) Spray fixture.

2. TEST PROCEDURE:

   (a) Subject the truck to a simulated rainfall of not less than 4 inches per hour. The simulated rainfall shall impinge on the truck at an angle of 45 degrees, plus or minus 5 degrees, with a force equivalent to a 30 mile per hour wind. The simulated rainfall shall disperse uniformly over the surfaces of the truck.
   (b) Expose the top and left side of the truck to the rain described above for a period of 5 minutes with engine off. With rain still flowing, start engine and allow engine to idle for 2 minutes.
   (c) Repeat (b) with the top and front exposed.
   (d) Repeat (b) with top and right exposed.
   (e) Repeat (b) with top and rear exposed.
   (f) Stop the rainfall and allow the truck to stand for not less than 30 minutes with the engine off. Drain a small quantity of fluid from each fluid reservoir (except cooling system) and examine for presence of water.
   (g) Check electrical components, gages, cab, and the air cleaner for detrimental water entry and damage.

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APPENDIX

TEST METHOD NO. 28

LOW-TEMPERATURE TEST (MINUS 25° F)

1. TEST APPARATUS:

   (a) Cold chamber.
   (b) Arctic clothing.
   (c) Thermocouples with potentiometer(s) or other temperature measuring device.
   (d) Battery hydrometer.
   (e) Wattmeter or ammeter.
   (f) Voltmeter.

2. TEST PROCEDURE:

   (a) Service the truck with applicable lubricants for minus 25° F operation including engine oil, transmission oil, drivetrain lubricants, hydraulic oil and antifreeze.
   (b) Check the specific gravity of battery electrolyte to insure full charge.
   (c) Subject the truck to an ambient temperature not greater than minus 30° F for a period of 6 hours without solar radiation. During and after this period, examine the truck for damage. Subject the truck to an ambient temperature of minus 25° F, or lower, until the engine coolant, fuel, and lubricating oil temperatures are stabilized from thermocouples located as follows:

      (1) In the two center cells of each battery between the plates and at middepth of the electrolyte.
      (2) In the volumetric center of the hydraulic system oil reservoir.
      (3) In the engine oil sump.
      (4) In the engine coolant in the engine cooling jacket discharge pipe.
      (5) In the chamber at a horizontal angle of 45 degrees from the sides and ends of the truck, not less than 3 feet from each corner of the truck at engine crankshaft level.

   (d) Maintain minus 25° F, or lower ambient temperature throughout the entire test.
   (e) One operator wearing arctic clothing, including mittens, shall perform all priming and starting operations.
   (f) Substitute or supplementary batteries or other electrical cranking energy shall not be used.

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APPENDIX

TEST METHOD NO. 28 (CONT'D)

(g) Glowplugs or other accessories working off the cranking batteries and auxiliary fluid priming systems which are permanently mounted on the truck and normally used in starting the engine may be used.

(h) Start the truck. Starting should be accomplished within 5 minutes of the initial attempt. Smooth engine running without unnatural or continued control manipulation, and without use of primers and priming fluids, should be obtained within 15 minutes after starting.

(i) When started, run engine 15 minutes. After completion of the 15 minute operation, operate the engine as follows:

(1) 10 minutes idling.
(2) 10 minutes half throttle.
(3) 5 minutes full throttle.

Perform the 25 minute cycle three times.

(j) Record temperatures prior to engine starting, immediately after engine starting, and at 10 minute intervals during warmup and cycling, including but not limited to the following:

(1) Battery.
(2) Hydraulic tank.
(3) Engine oil.
(4) Engine coolant.
(5) Chamber air.

(k) During engine cycling, and while wearing arctic mittens and boots, the operator shall perform each of the following:

(1) Engage and disengage power train to the wheels in all gears.
(2) Operate through two complete movement cycles all brakes, top handler operations, shafts and cylinders (as permitted by space limitations in chamber).
(4) Mount to and dismount from the operator's position on the truck.

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APPENDIX

TEST METHOD NO. 29

PERFORMANCE (FORDING)

1. TEST COURSE:

(a) The test course shall be a dry land area and a body of water of sufficient depth to permit submersion of the truck to minimum 60 inch depth.

2. TEST PROCEDURE:

(a) Paint a horizontal white line on each side of the truck at the 60 inch fording depth.
(b) Fluid samples shall be taken from the lowest point of the following at the beginning of the test, and not less than 10 minutes after conclusion of the test:

(1) Torque converter sump.
(2) Engine oil sump.
(3) Transmission.
(4) Transfer case.
(5) Fuel tank.
(6) Hydraulic reservoir.
(7) Each axle.

Samples shall be replaced with fresh lubricant. Sample size shall be not less than 200 cubic centimeters each. Samples shall be analyzed for volumetric water content by a means having an accuracy of within plus or minus 1 percent.

(c) Operate the truck as follows:

(1) Drive the truck into the water to the specified depth. Maneuver the truck for a period of not less than 15 minutes in the water as follows:

(aa) Drive the truck in forward and reverse at minimum of 3 mph for a period of 12 minutes. Stop the truck by applying the service brakes for each change of direction.
(bb) During the 12 minute cycle in (aa) the steering system shall be actuated from stop to stop.
(cc) At conclusion of the 12 minutes of operation shut off the engine. Restart the engine after 3 minutes and drive out of the water.
(dd) Allow truck to set for 15 minutes on dry land with engine off.
(ee) Repeat the procedures (aa) through (dd) four additional times for a total of not less than 2-1/2 hours.

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APPENDIX

TEST METHOD NO. 30

TOWING

1. TEST COURSE:

   (a) Prepared or semi-prepared surface.

2. TEST APPARATUS:

   (a) Towing vehicle with pintle hook and speed indicator.
   (b) Towbar in accordance with MS500048, heavy duty capacity.

3. TEST PROCEDURE:

   (a) Attach towbar to the towing lugs.
   (b) Prepare the truck for towing. Record the time and procedures.
   (c) Tow the truck for not less than 10 miles at an average speed of 4 mph. As a minimum while towing the truck, negotiate two 180 degree turns. Measure diameter of the 180 degree turns. Observe tracking ability and note any loss of control.

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APPENDIX

TEST METHOD NO. 31

CONTROLS

1. TEST APPARATUS:

   (a) Thermocouples and potentiometer or equivalent temperature measuring device.
   (b) Force measurement equipment.
   (c) Linear measurement equipment.

2. TEST PROCEDURE:

   (a) Install the thermocouple in the hydraulic reservoir.
   (b) With an oil temperature of 140° F to 180° F and the engine operating at no load governed speed, measure the force and distance required to shift each hydraulic control valve at the handgrip to each position.
   (c) Apply 100 pounds of force to each valve lever in each shifting direction and maintain the force for not less than 15 seconds.
   (d) Apply 100 pounds of force to each transmission control lever in each direction and maintain the force for not less than 15 seconds.

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APPENDIX

TEST METHOD NO. 32

REAR AXLE OSCILLATION

1. TEST APPARATUS:

   (a) Clinometer.
   (b) Crane to lift back of the truck.

2. TEST PROCEDURE:

   (a) Raise the truck, keeping the truck horizontal, to sufficient height to allow the rear axle to oscillate from stop to stop.
   (b) Attach the clinometer to the rear axle.
   (c) Oscillate the rear axle fully clockwise. Measure and record the angle from the horizontal.
   (d) Oscillate the rear axle fully counterclockwise. Measure and record the angle from the horizontal.

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APPENDIX

TEST METHOD NO. 33

TRUCK OPERATIONS

1. TEST COURSE:

   (a) The test course shall be a level, improved unpaved surface laid out typically in accordance with Figure A-2. Station 2 and Station 3 shall be minimum 8 foot high containers capable of supporting fully loaded 20, 35 and 40 foot containers. Station 1 shall be ground level.
   (b) Obstacles shall be in accordance with Figure A-3 and placed such that the nearest obstacle is not less than 30 feet from Station 1.

2. TEST APPARATUS:

   (a) Two 20 foot, two 35 foot and two 40 foot containers with rated load.
   (b) Two 20 foot, two 35 foot and two 40 foot containers for Stations 2 and 3.
   (c) Thermocouples to measure:

      (1) Engine coolant at engine outlet to the radiator.
      (2) Engine coolant from the radiator to the engine.
      (3) Transmission sump.
      (4) Engine oil sump and cooler.
      (5) Drive axle differential.
      (6) Hydraulic oil reservoir.

   (d) Thermometer to measure ambient air during test.

3. TEST PROCEDURE:

   (a) Initially place rated loads on containers at Stations 1 and 3.
   (b) Pick up container from Station 1, back up truck being sure all wheels travel over obstacles, travel forward and deposit load at Station 2.
   (c) Disengage load at Station 2, back out, then travel forward to Station 3. Prior to engaging load at Station 3, raise mast to maximum lift height to activate relief valve, lower top handler and engage load.
   (d) Pick up load at Station 3, back out, then travel forward and deposit load at Station 1 being sure all wheels travel over obstacles.
   (e) Disengage load at Station 1, back out being sure all wheels travel over obstacles, then travel forward to Station 2.
   (f) Engage load at Station 2, back out, then travel forward and deposit load at Station 3.
   (g) Disengage load at Station 3. Back out and travel forward to Station 1.

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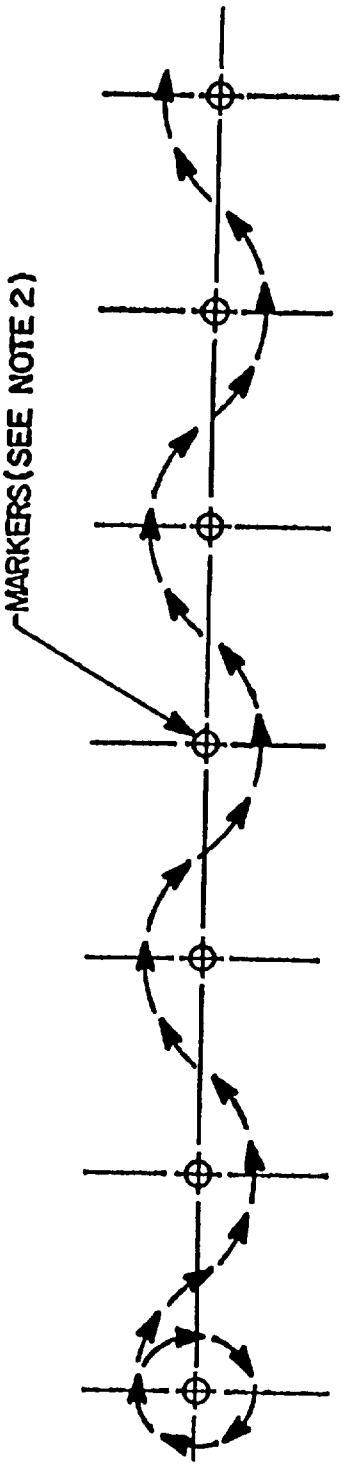
APPENDIX

TEST METHOD NO. 33 (CONT'D)

(h) Repeat 3(b) through 3(g).
(i) Steps 3(b) through 3(h) shall constitute 1 cycle. This includes 6 container handlings and 4 trips around the circle.
(j) For 50 percent of the cycles, Station 2 and Station 3 shall be interchanged such that they are in clockwise position in lieu of initial counterclockwise. Stations shall be changed at the end of each 8 hour operating day.

(k) Record total cycles for each hour of operation.
(l) Record each temperature for each hour of operation for one 8 hour shift each week of operation.
(m) Change tophandler and container sizes every 450 cycles of operation.

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NOTES:

1. THE SPACE BETWEEN THE MARKERS SHALL BE EQUAL TO TWO TIMES THE CURB CLEARANCE CIRCLE DIAMETER.
2. MARKERS SHALL BE HIGH ENOUGH TO BE EASILY SEEN BY THE OPERATOR WHILE GOING THROUGH THE TEST COURSE.

FIGURE A-1. Obstacle course no.1 for steering test.

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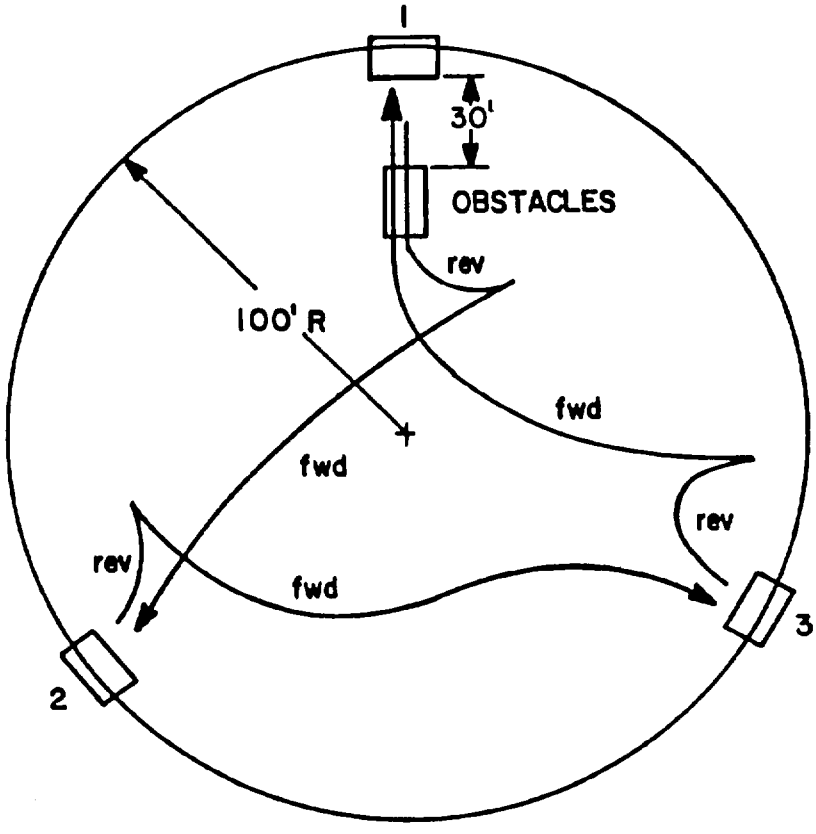
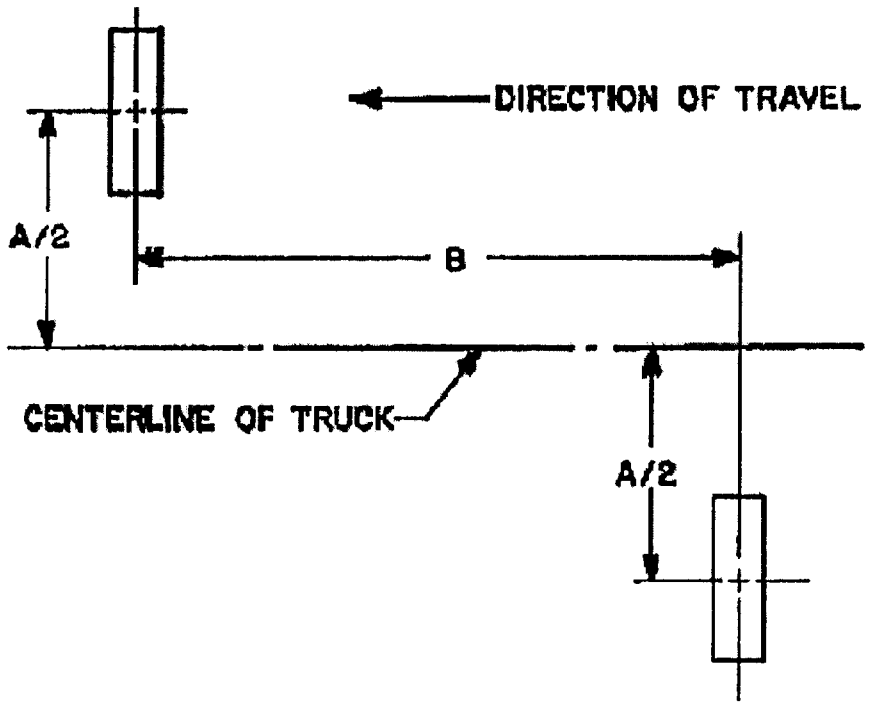


FIGURE A-2. Truck operations course.

X-3364D

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LAYOUT OF OBSTACLE COURSE



A= DISTANCE BETWEEN CENTERLINES OF DRIVING WHEELS.
B= WHEEL BASE OF TRUCK PLUS 2 FEET.

BLOCKS SHALL BE 8 INCHES HIGH X 8 INCHES WIDE X 6 FEET LONG.

FIGURE A-3. Details of obstacle construction.

X-3365

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-52951(ME)	2. DOCUMENT DATE (YYMMDD) 780417

3. DOCUMENT TITLE
TRUK, CONTAINER HANDLER, WITH TOP LIFT ATTACHMENTS, DIESEL ENGINE DRIVEN, PNEUMATIC TIRED, FOUR WHEEL DRIVEN, ARTICULATED FRAME STEER, ROUGH TERRAIN; 50,000 POUND CAPACITY AT 48 INCH LOAD CENTER.

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial DSN (If applicable)	7. DATE SUBMITTED (YYMMDD)

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c. ADDRESS (Include Zip Code) US Army Tank-automotive and Armaments Command Attn: AMSTA-TR-E-ELE/21 Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 681-9340 DSN 761-9340

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